Panoramic Radiography

Panoramic Radiography

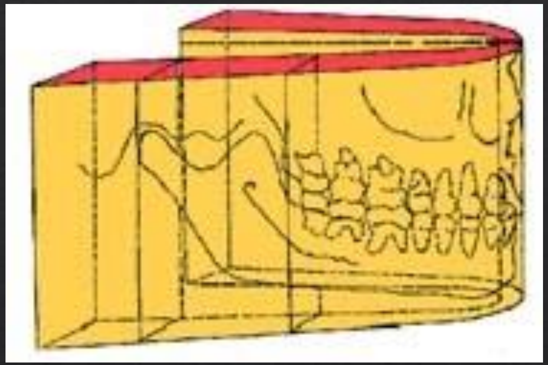
- Panoramic radiography is an extra-oral radiographic projection which produces an image of a curved 3D layer. A sharply depicted image layer or focal trough is created as a result of the simultaneous movement of a highly collimated source of radiation and a flat or curved film/sensor contained inside a cassette.

History

- Drs. Paatero and Numata independently were the first to describe the principles of panoramic radiography.
- Dr. Numata in 1933 was the first to propose development of extraoral source of radiation and placed a curved film in the mouth lingual to the teeth and used a slit beam of radiation which rotated around the patient's jaw to expose the film.
- Dr. Paatero builds his first working prototype (Parablograf) in 1947 based on his invention.

Panoramic Radiography

- Pantomography
- Body section imaging technique that results in a wide, curved image layer depicting the maxillary and mandibular dental arches and their supporting structures
- Structures are flattened and spread out



Normal panoramic image

Panoramic imaging-Indications

- Overall evaluation of dentition
- Examine for intraosseous pathology, such as cysts, tumors, or infections
- Gross evaluation of temporomandibular joints
- Evaluation of position of impacted teeth
- Evaluation of eruption of permanent dentition
- Dentomaxillofacial trauma
- Developmental disturbances of maxillofacial skeleton

Advantages	• Broad coverage of facial bones and teeth • Low radiation dose • Easy panoramic technique • Suitable for patients with trismus or who cannot tolerate intraoral radiography • Quick and convenient • Useful for patient education and case presentations

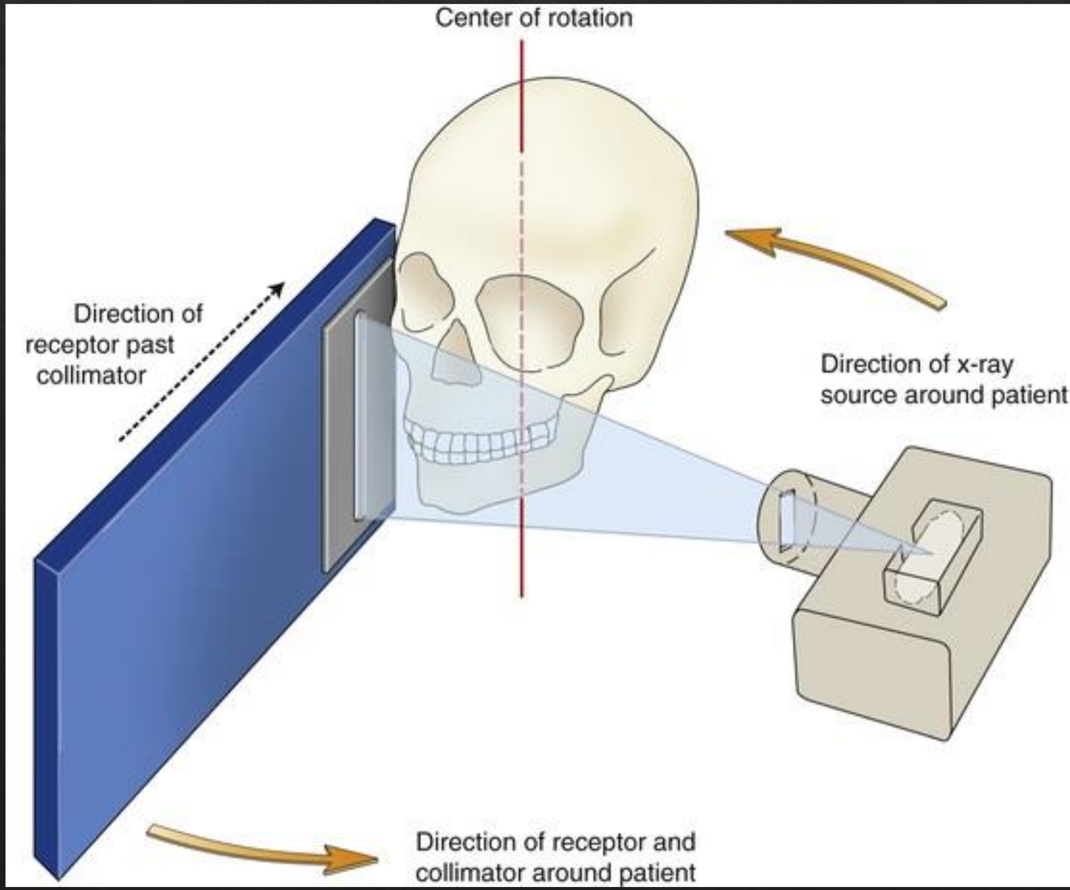
Disadvantages	• Lower-resolution images compared to intraoral radiographs • Unequal magnification — unreliable linear measurements • Superimposition of real, double, and ghost images • Requires accurate patient positioning to avoid errors • Difficult imaging in severe maxillomandibular discrepancies

- The availability of a panoramic radiograph for an adult patient often does **not** eliminate the need for intraoral films for diagnosing common dental diseases.
- When a **full-mouth series** is already available, a simultaneous panoramic radiograph typically adds **little or no additional diagnostic value** for general dental care.
- A **panoramic radiograph combined with bitewings**, or a combination of **bitewings and selected periapical films**, can provide diagnostic information **comparable** to a full-mouth series.
- **Problems associated with panoramic radiography** include:
  - Unequal magnification and **geometric distortion** across the image.
  - **Overlapping structures** (e.g., cervical spine) can obscure odontogenic lesions, especially in the **incisor region**.
  - Clinically important structures located **outside the focal trough** may appear **distorted or may not be visible at all**.

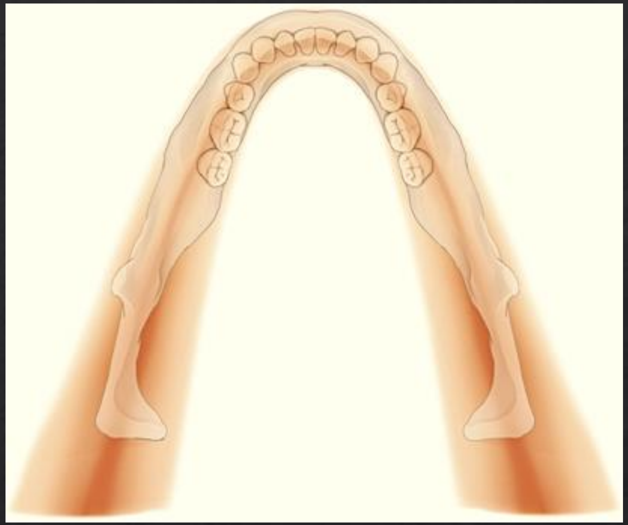
Principles of imaging formation

General scheme of panoramic acquisition

- X-ray source and receptor rotate around patient's head
- X-ray are collimated –narrow vertical beam at the source and receptor
- Receptor moves past the collimator during exposure cycle
- Acquires sequential vertical images



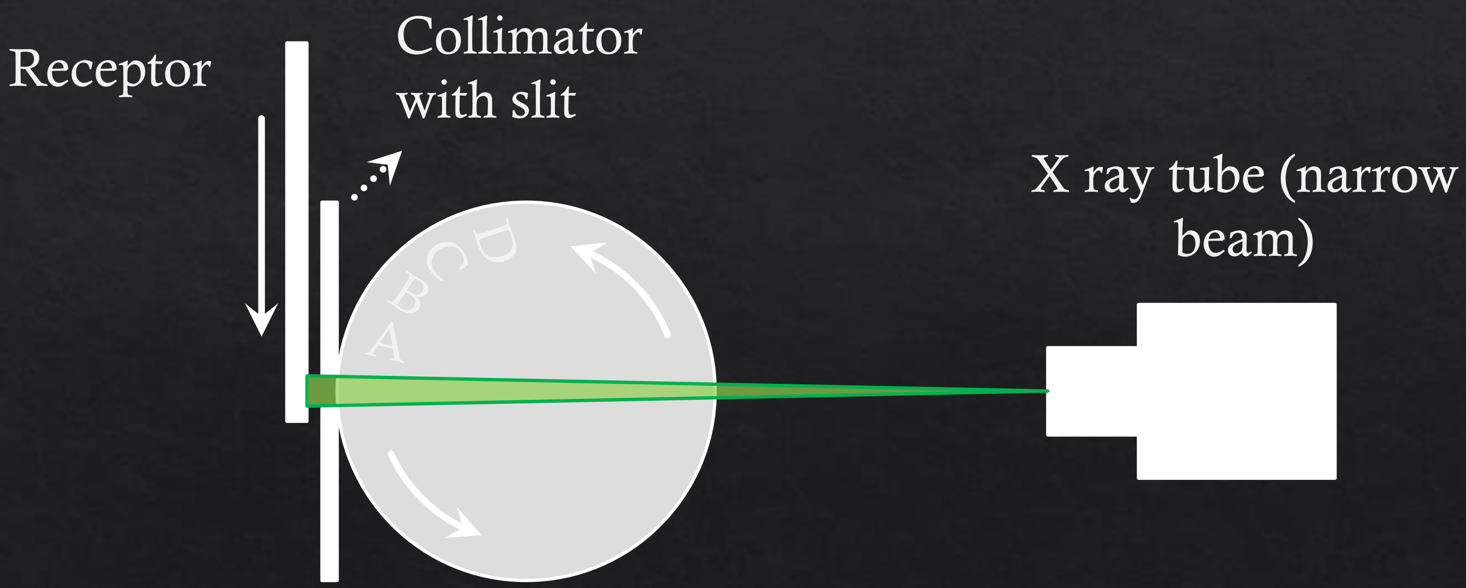
Key concept-Image layer (Focal Trough)



- The moving source and receptor generate a zone of image sharpness-focal trough
- Focal trough or image layer is a wide, 3D curved zone, where the structures positioned within this zone are reasonably well defined on the panoramic image
- Structures positioned in the center of the focal trough are the clearest and those that are progressively farther from the center of the focal trough become progressively less clear

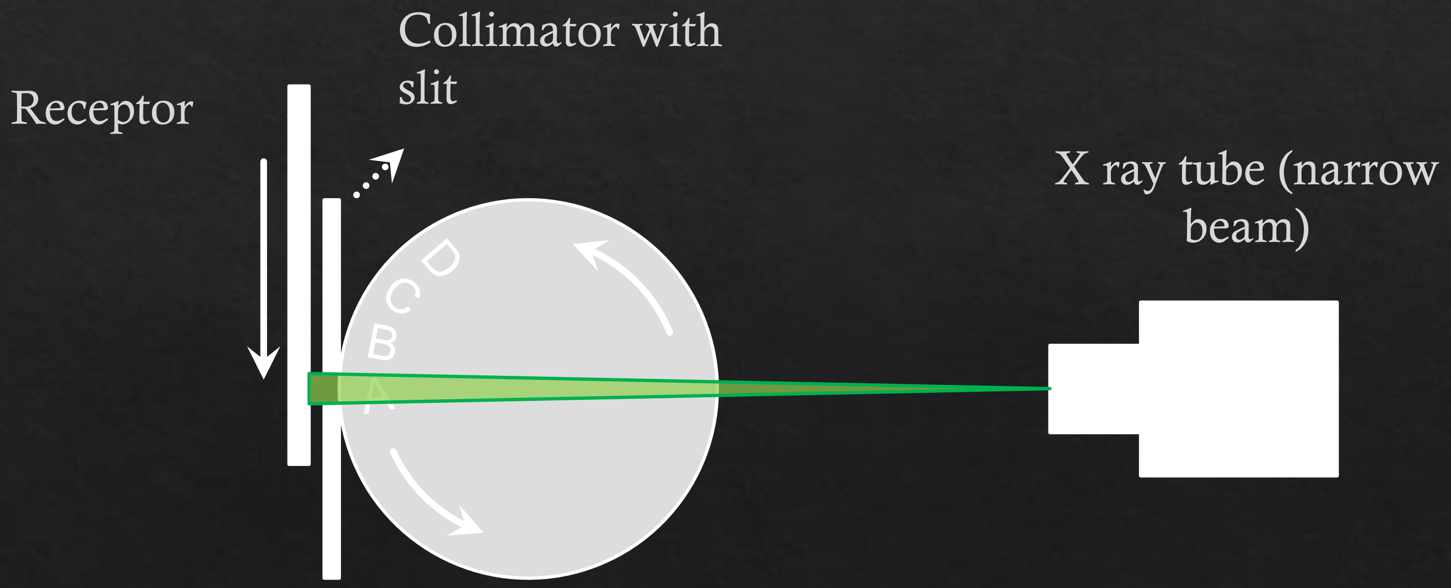
Panoramic Image Acquisition

- Imagine a rotating disc exposed with a stationary narrow X ray beam, and an image receptor moving (at the same speed as objects A-D on the rotating disc) along a narrow slit in a collimator



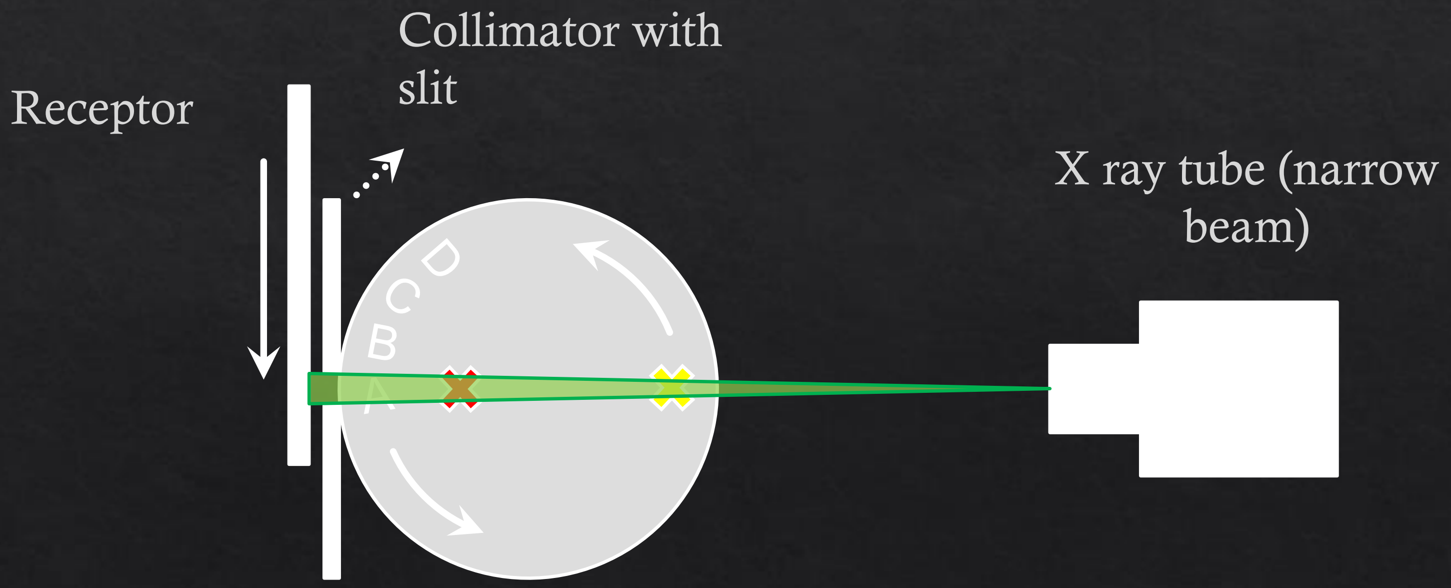
Panoramic Image Acquisition

- When objects A-D pass along the slit, they will appear on the image receptor sharply (since they are moving at the same speed and in the same direction as the receptor)



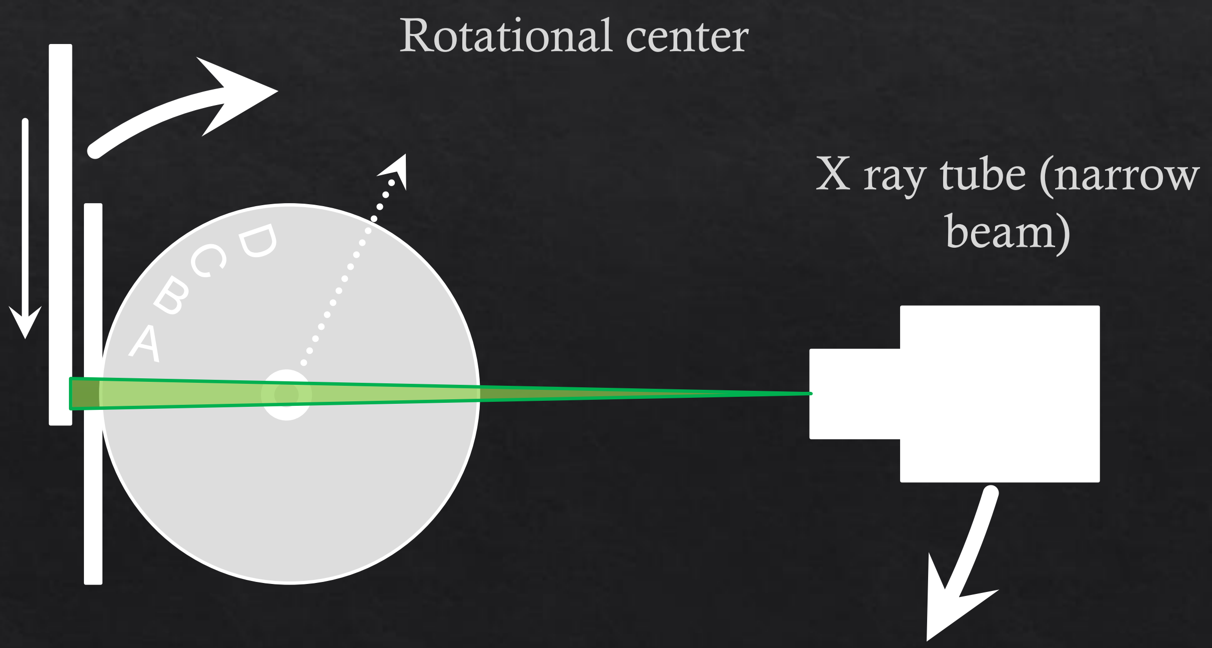
Panoramic Image Acquisition

- Objects closer to the X ray tube are moving at slower speeds (✖) or in opposite direction to the receptor (✖) and will therefore appear blurry on the receptor



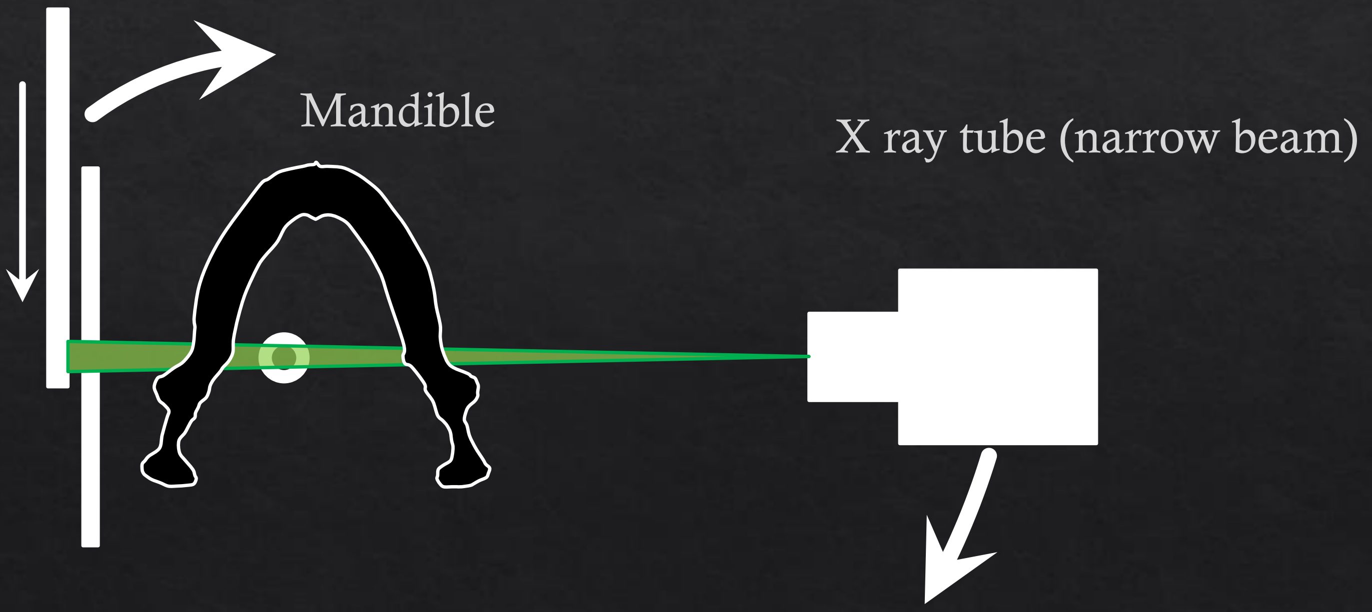
Panoramic Image Acquisition

- The same principle applies if the disc is stationary and, instead, both the X ray tube and collimator-receptor rotate around the centre of the disc (◉) (while the receptor is moving along the slit)



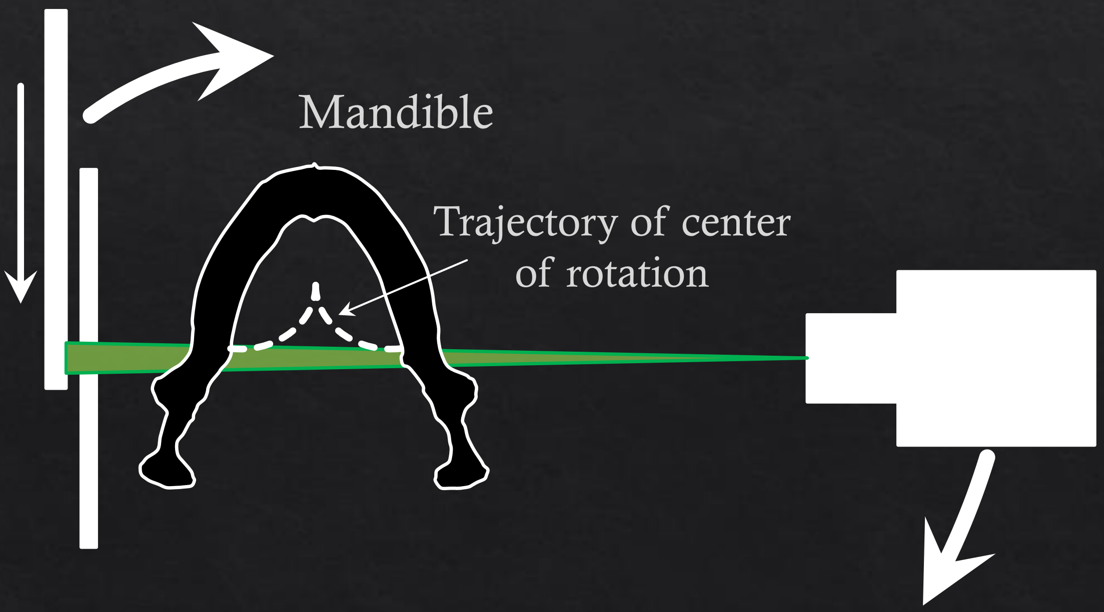
Panoramic Image Acquisition

- If a patient is exposed using this set-up, the translational speed of the receptor and the centre of rotation can be determined in order to sharply visualize the jaw bones



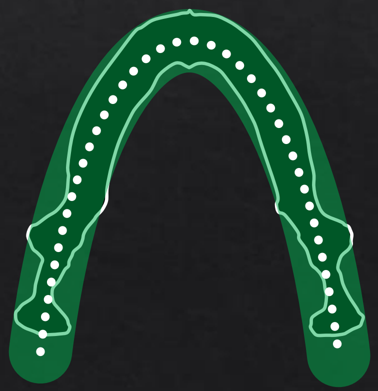
Panoramic Image Acquisition

- Due to the shape of the human jaw bones, the rotational centre should not be fixed
- Modern panoramic equipment use a continuously moving centre of rotation (----) to ensure optimal sharpness along the jaw arch



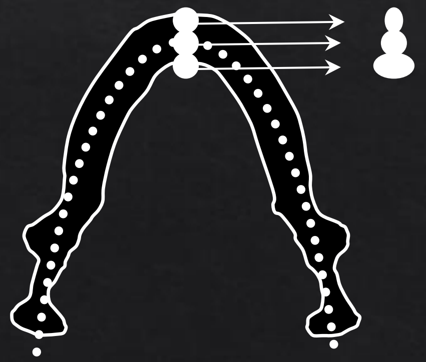
Panoramic Image Acquisition

- The result is a panoramic radiograph in which only structures along the dental arch are in focus
  - ‘Focal trough’ or ‘image layer’ (·····)
  - Different rotation paths available for ≠ jaw shapes
  - The closer a structure is to the centre of the trough (·····), the sharper it will appear



Magnification in panoramic radiography

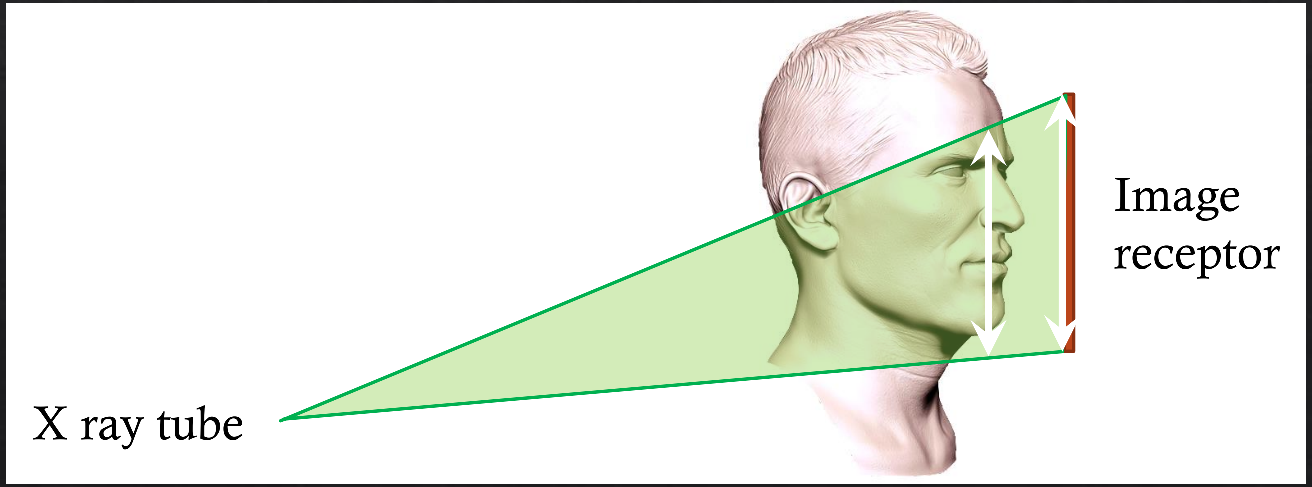
- Image distortion is inherent to the panoramic technique: linear/angular measurements not reliable
- Horizontal magnification
  - Depends on relative distance to focal trough
  - Different for anterior & posterior structures ('snowman principle')



- Middle of trough: no magnification
- Anterior: horizontal narrowing
- Posterior: horizontal widening

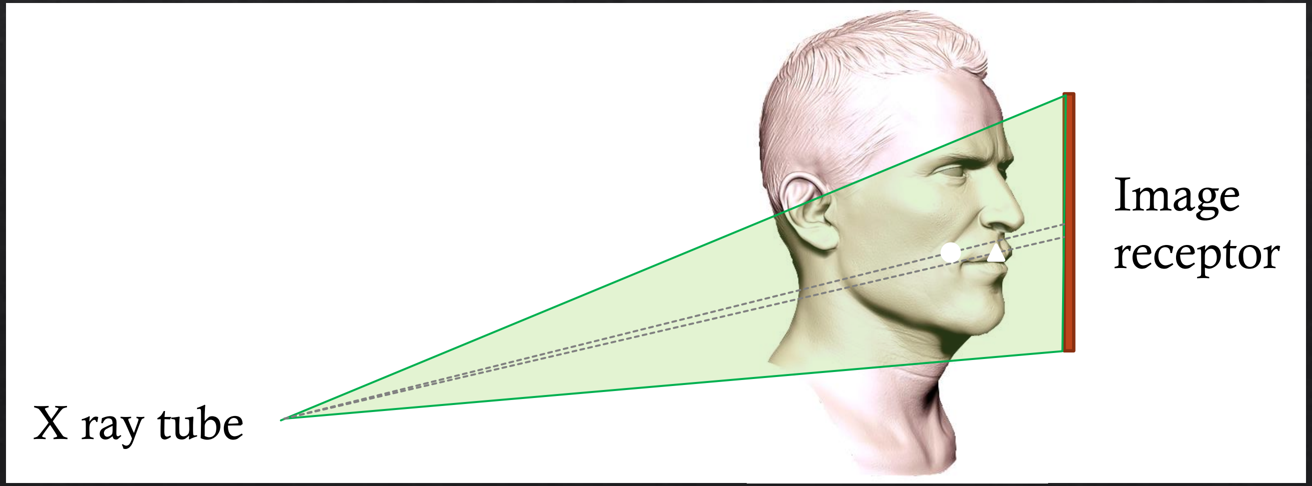
Magnification in panoramic radiography

- Vertical magnification
  - Depends on source-object distance
  - Distance can be constant for certain units
  - Slight difference in magnification between edges of focal trough (due to difference in distance to source)



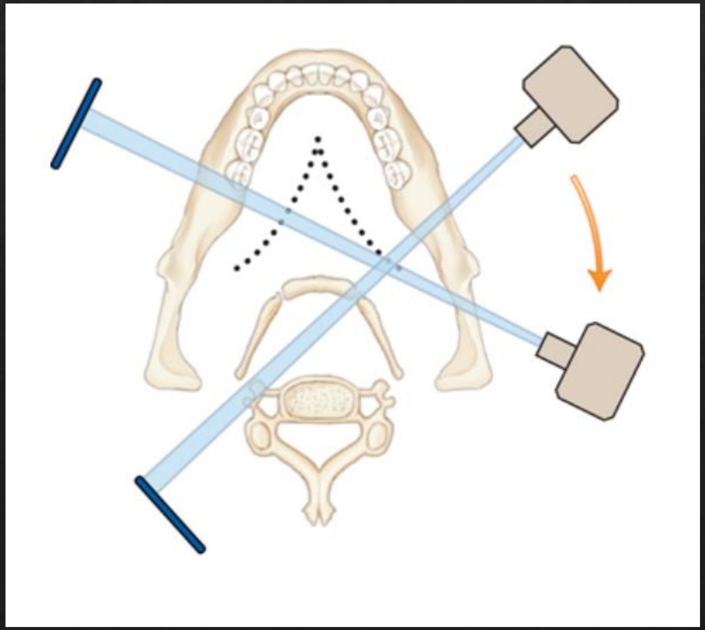
Magnification in panoramic radiography

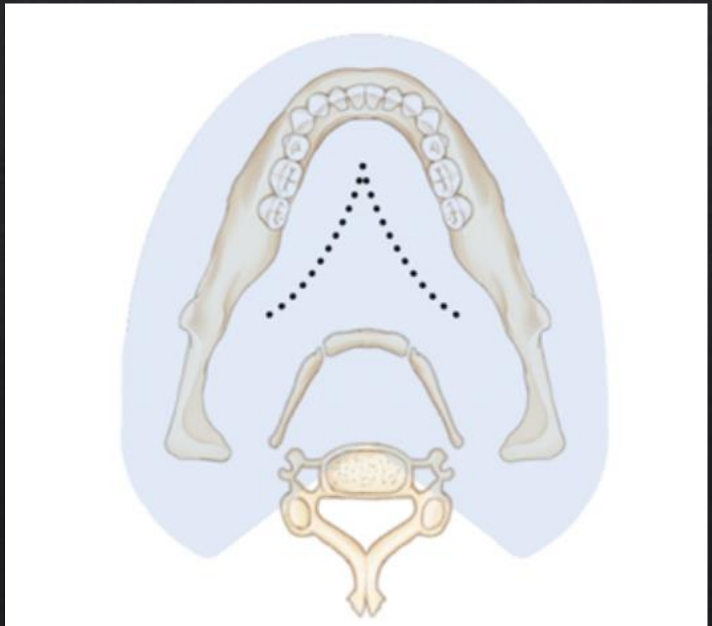
- Vertical distortion
  - Due to upward angle of X-ray beam
  - Structures closer to tube (i.e. lingually) (●) projected higher on image receptor than structures closer to receptor (▲)
    - e.g. cervical spine vs. mandible

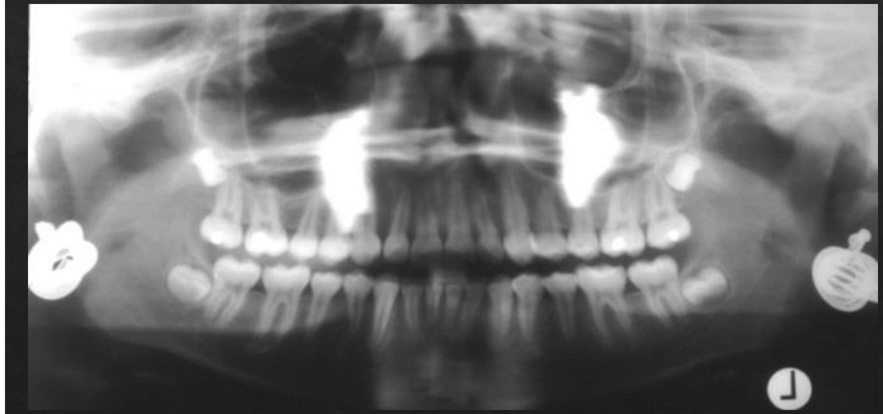


Real/Double/Ghost Images

- Real image
  - Object is between centre of rotation and receptor
  - Sharper if closer to focal trough

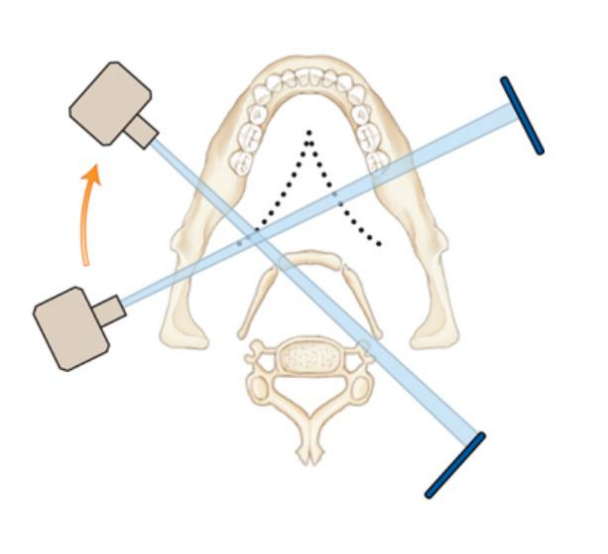


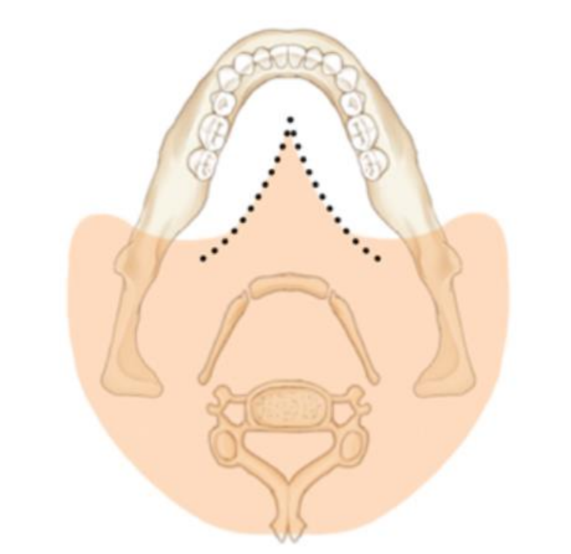


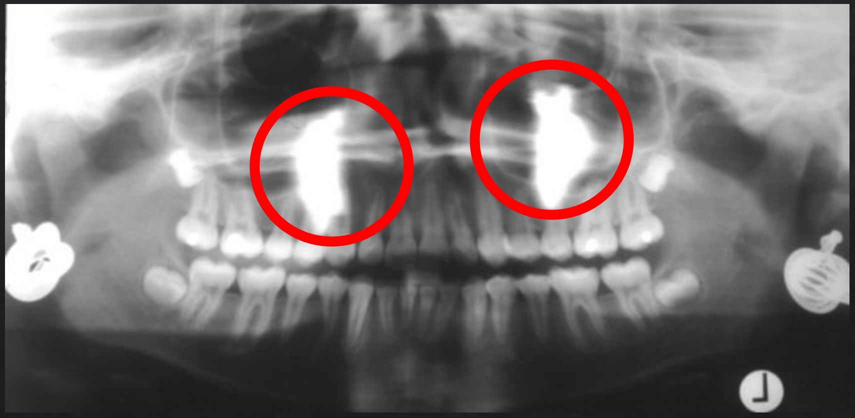


Real/Double/Ghost Images

- Ghost image
  - High attenuation object between x-ray source and centre of rotation
  - Appears on opposite side of image (L↔R) and higher on the image (due to upward beam) than their anatomical location
  - Far outside focal trough: appears blurred and magnified
  - Mandibular ramus, hyoid bone, cervical spine, earrings, hair pins, etc.

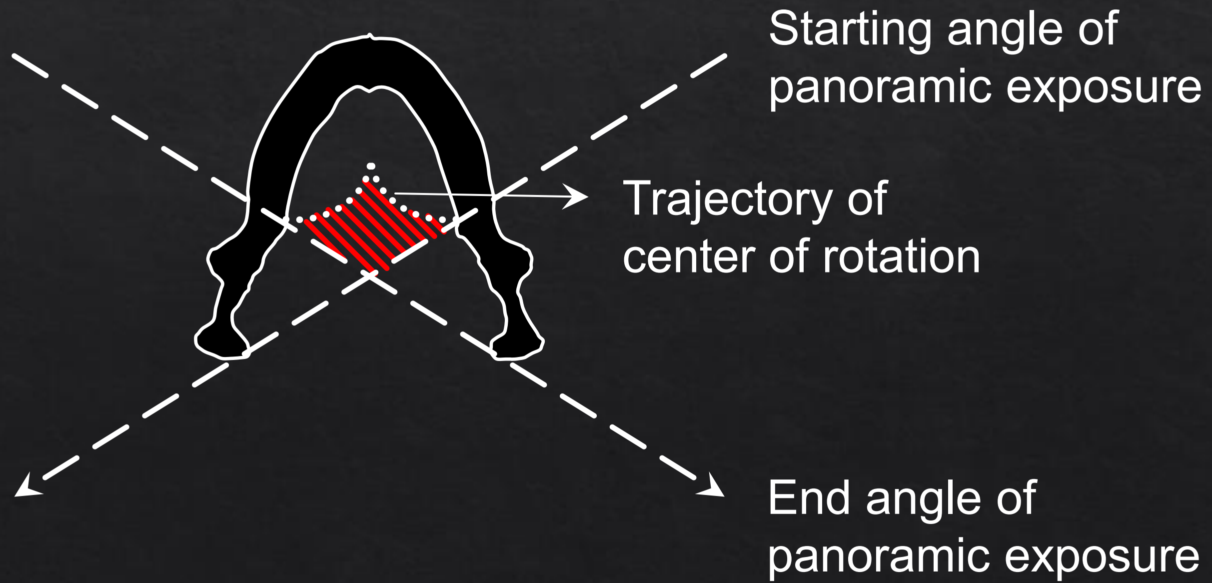






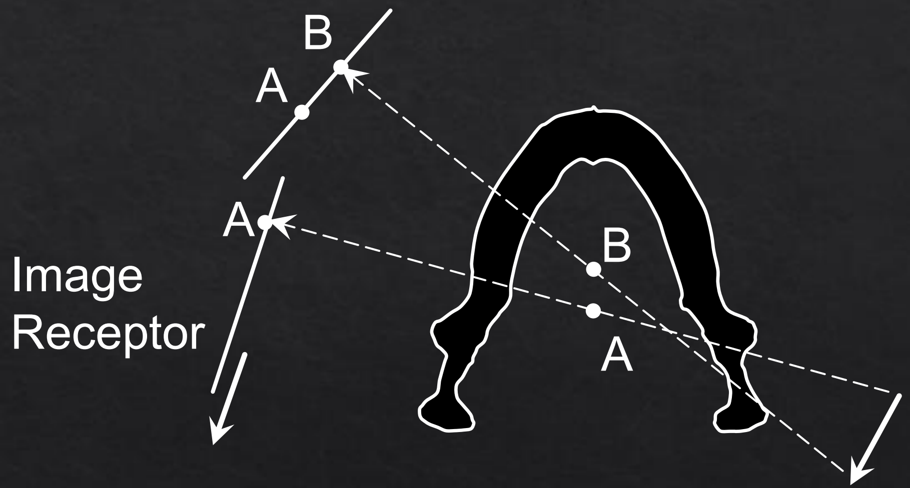
Real/Double/Ghost Images

- Double image: A double image is a pair of real images formed by an object lying within a zone that is intercepted twice by the beam
  - Object is posterior to centre of rotation and intercepted by beam twice (area marked by)
  - Hyoid bone, epiglottis, cervical spine, palate

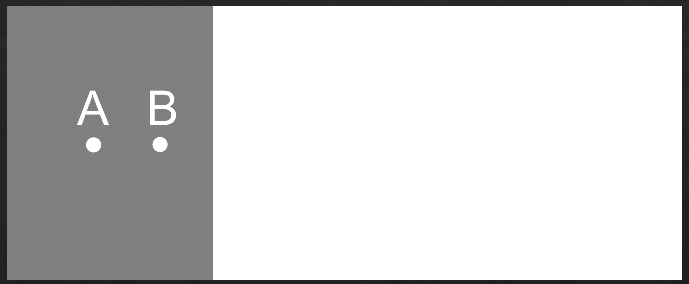


Real/Double/Ghost Images

- Double image
  - First projection of points A and B
    (A is projected first, then B)



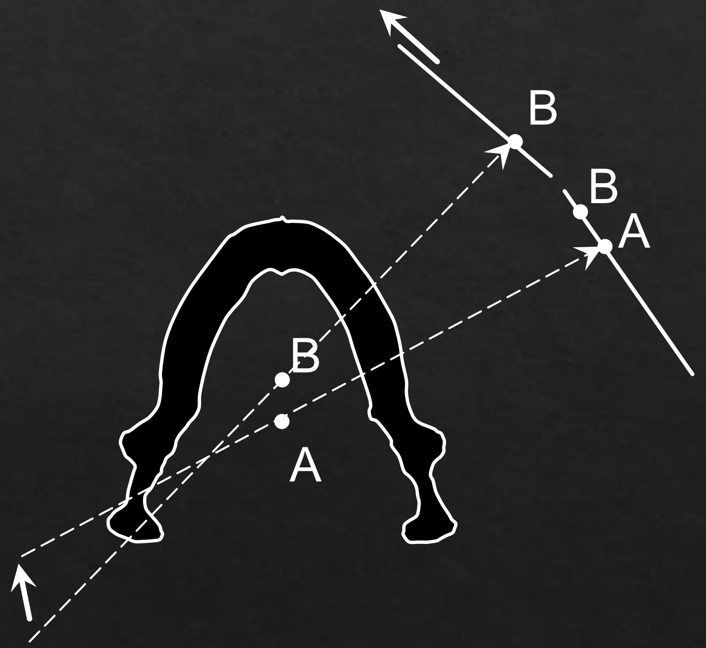
Panoramic image



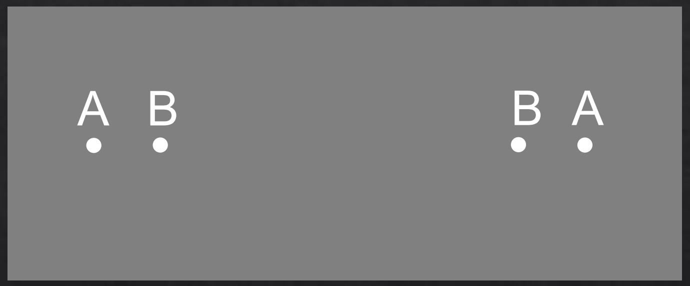
(exposed section of image
shown in grey)

Real/Double/Ghost Images

- Double image
  - Second projection of points A and B (B is projected first, then A)

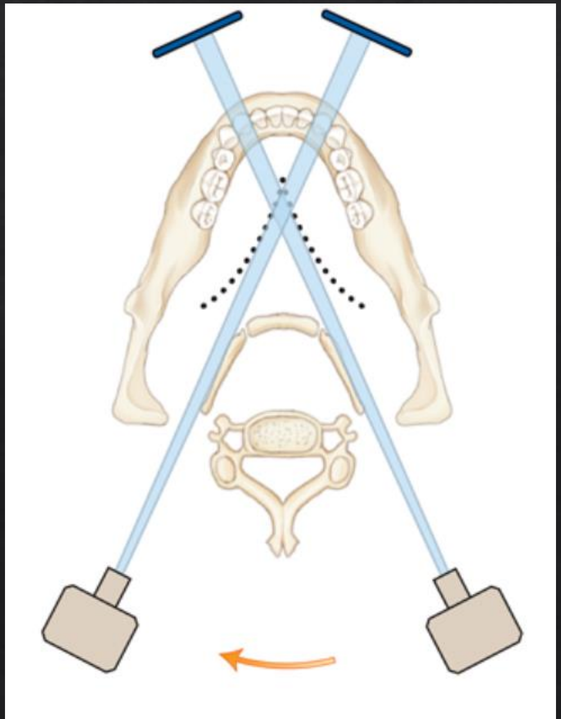


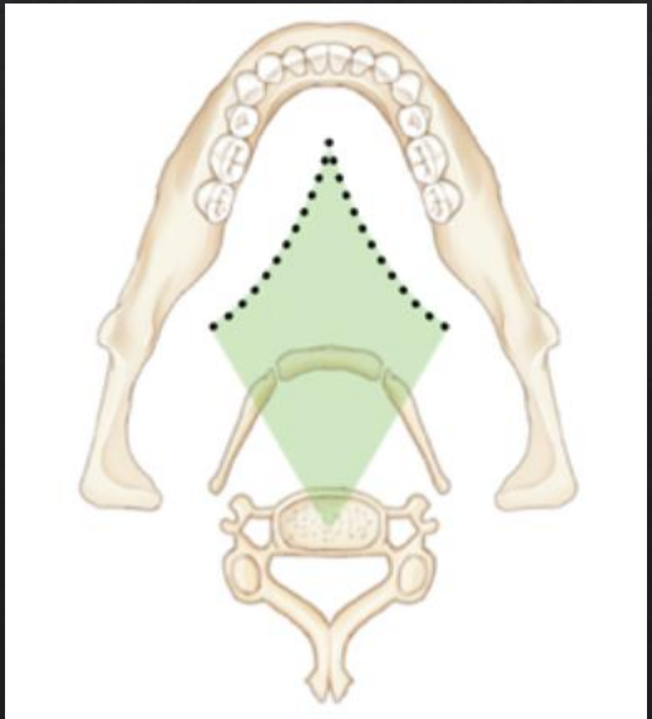
Panoramic image



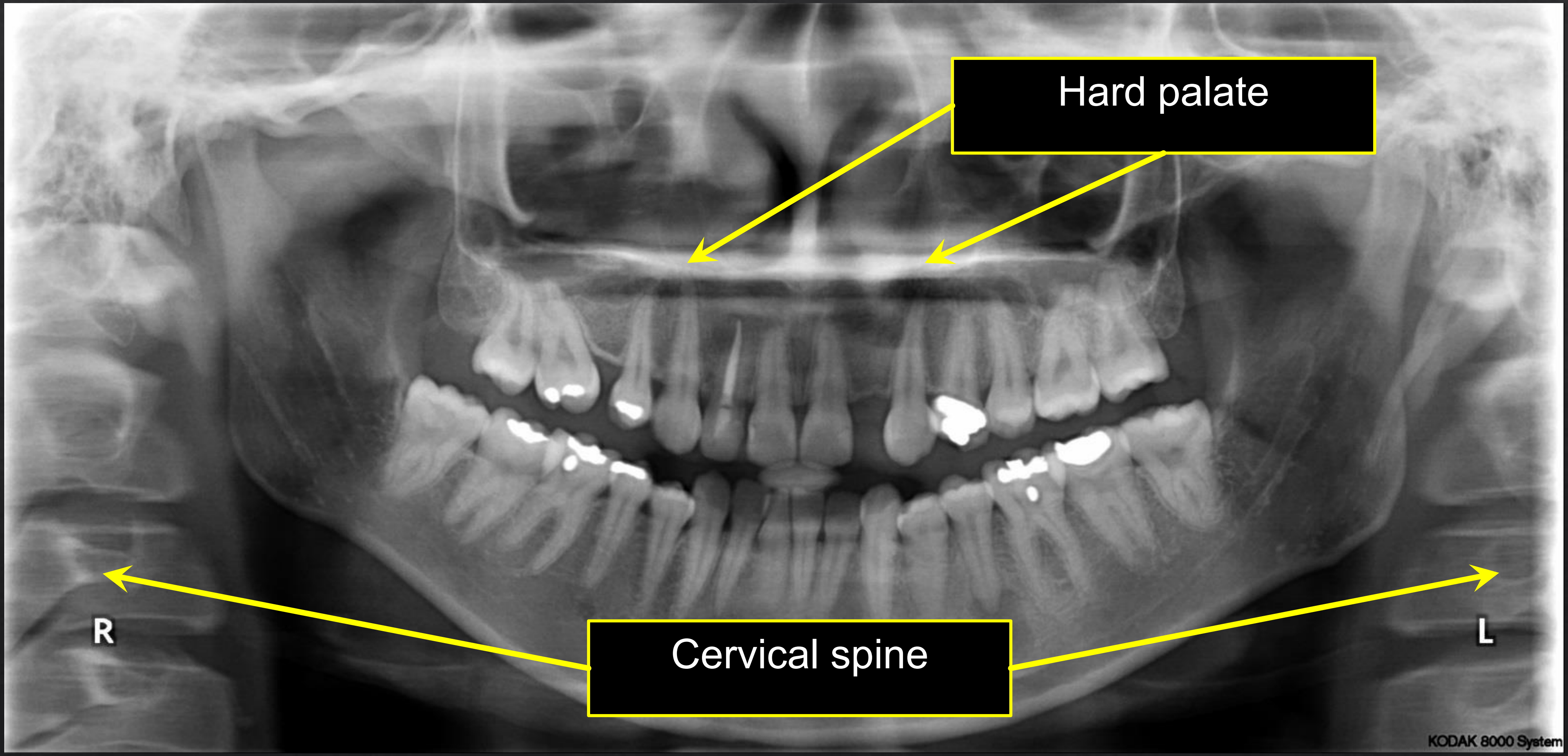
Real/Double/Ghost Images

- Double images are mirror images,
- Each image will appear on the same location but opposite sides
- Each image will have similar proportions
- Double images only occur with multiple objects falling in the diamond-shaped zone in the midline





Double image



Patient positioning

- Panoramic radiography is highly sensitive to correct patient positioning
  - Mandibular and maxillary structures should be positioned in the focal trough
  - Improper head orientation and tongue position can lead to shading and other artefacts

Patient positioning

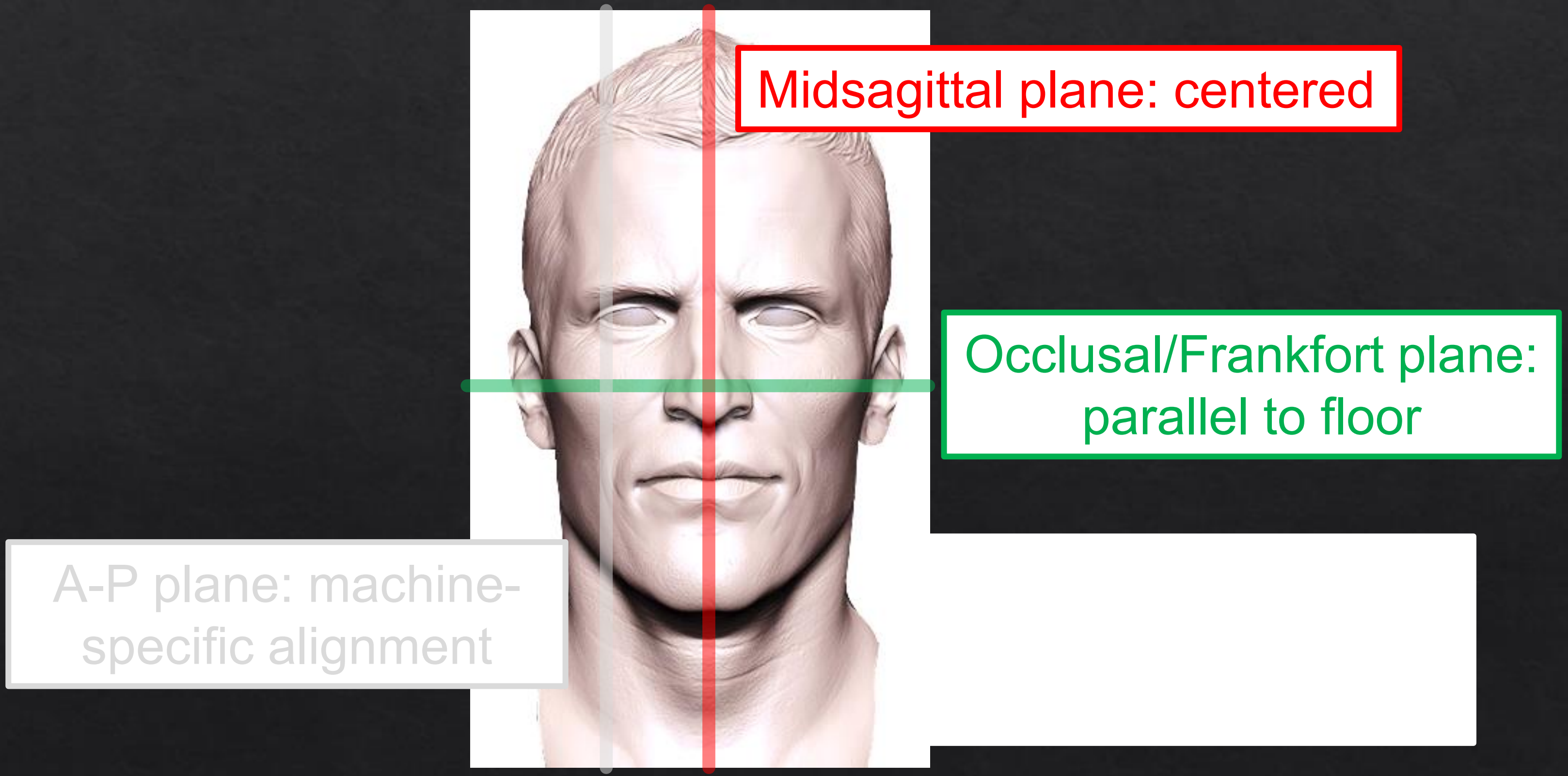
- Metal objects such as earring should be removed to avoid interference with image interpretation
- Possible collision of the tube/receptor with the patient's shoulder should be checked
  - "Test run" (no exposure) can be used as verification

Patient positioning

- Patient bites in bite block or uses chin rest
  - Corresponds with anterior part of focal trough to ensure coverage of anterior teeth (focal trough is thinnest in this region)

Patient positioning

- Head alignment done in three different planes (with help of alignment lights and mirrors)



Patient positioning

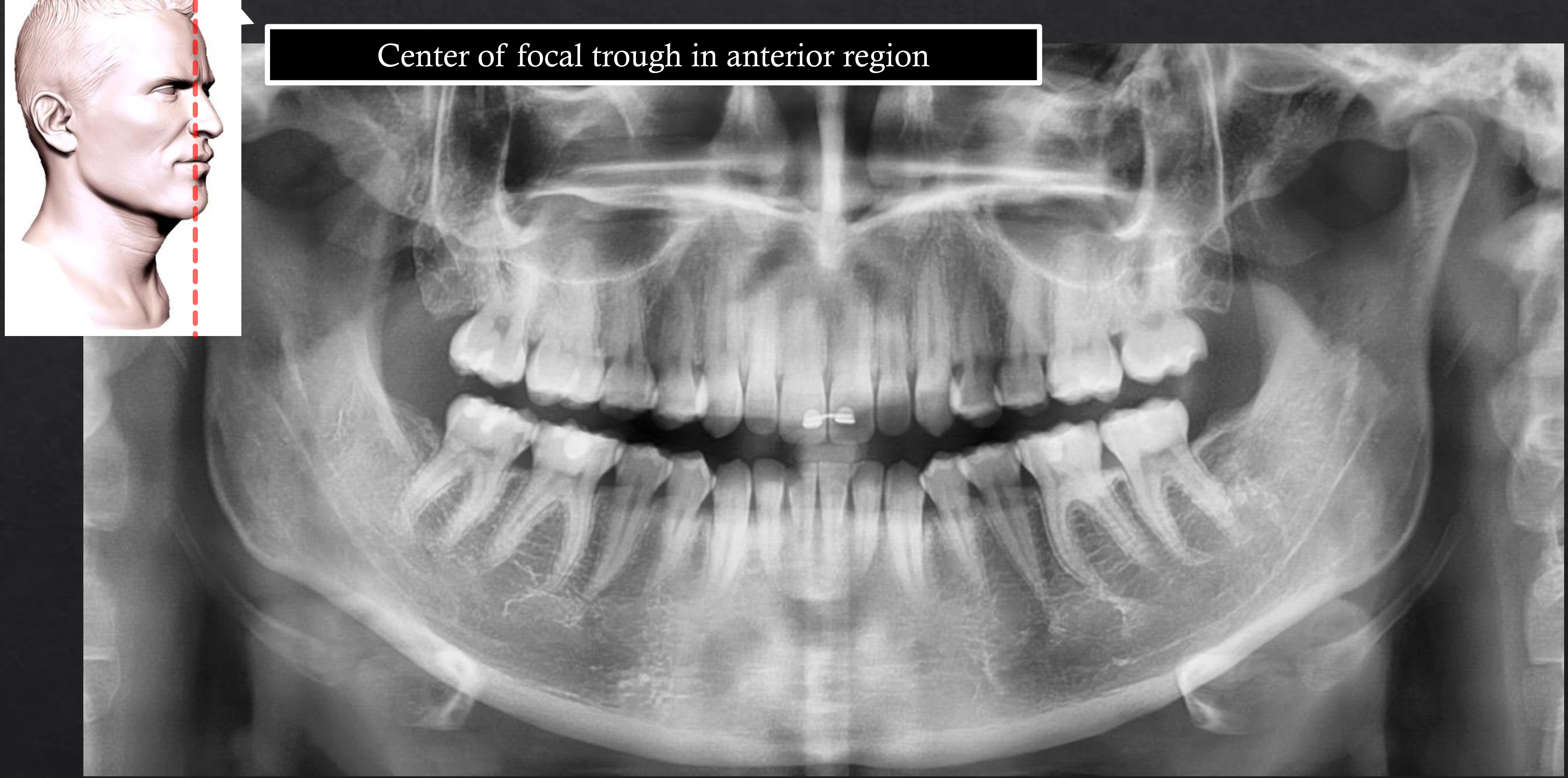
- Patient instructions before acquisition
  - No movement (~15-20s; fast modes available when needed)
  - Close lips, place and hold tongue against palate

Patient positioning

- Essential image characteristics
  - Edge to edge incisors
  - No removable metallic foreign bodies
  - No motion artifacts
  - Tongue is against roof of mouth
  - Minimization of spine shadow
  - No A-P positioning errors
  - No mid-sagittal plane positioning errors
  - No occlusal plane positioning errors
  - Correct position of spinal column
  - Good density and adequate contrast between enamel and dentine
  - Name/date/left-right markers visible

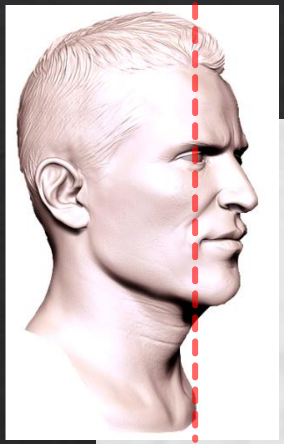
Patient positioning

Proper patient positioning

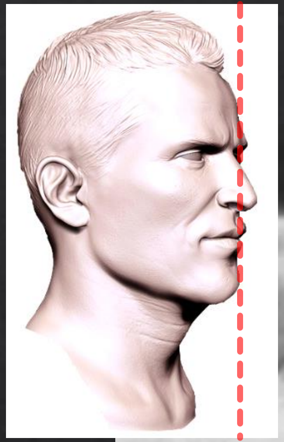


Patient positioning

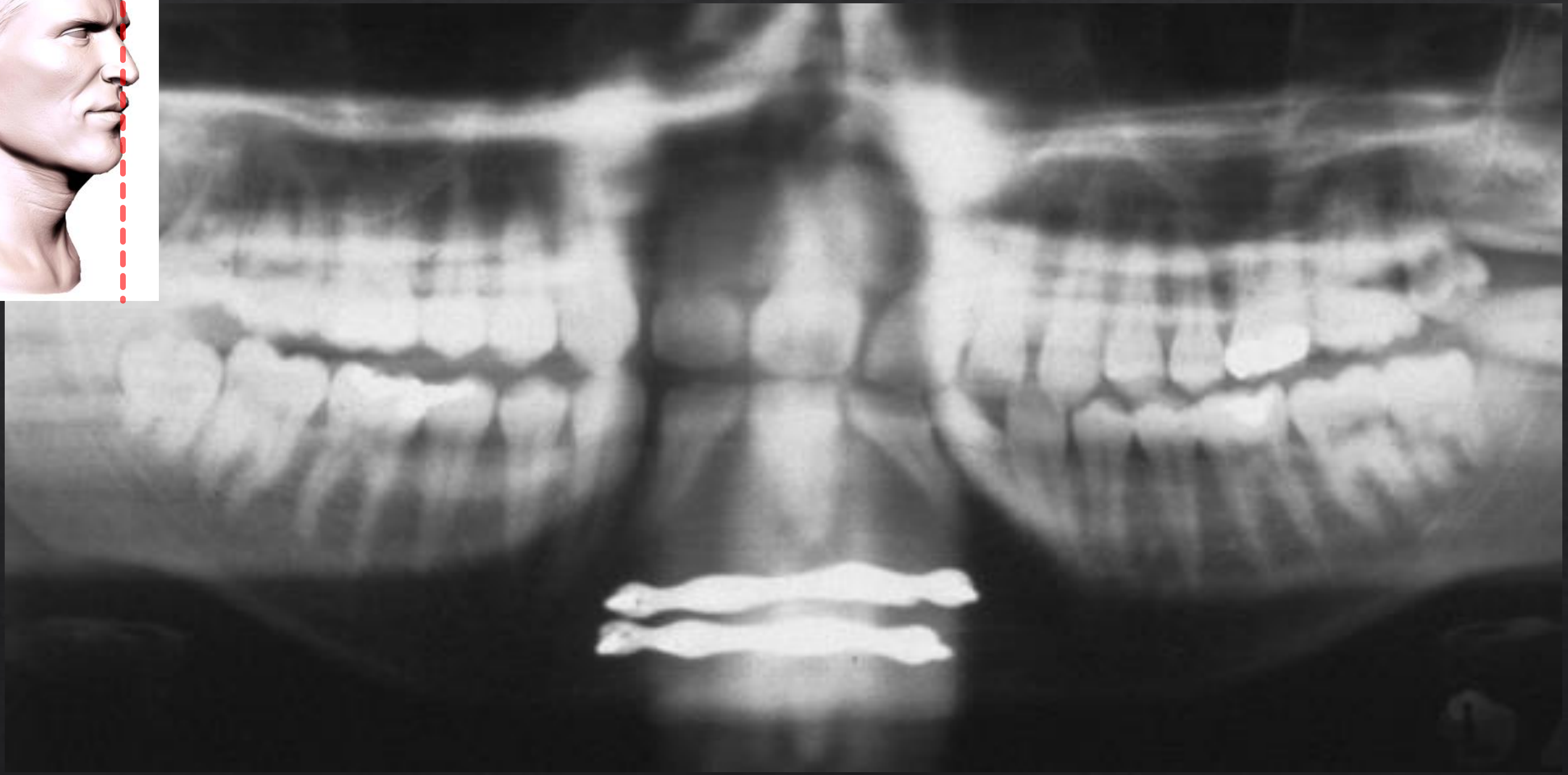
Head too much forward; 'minification'



Patient positioning



Head too much backward; magnification

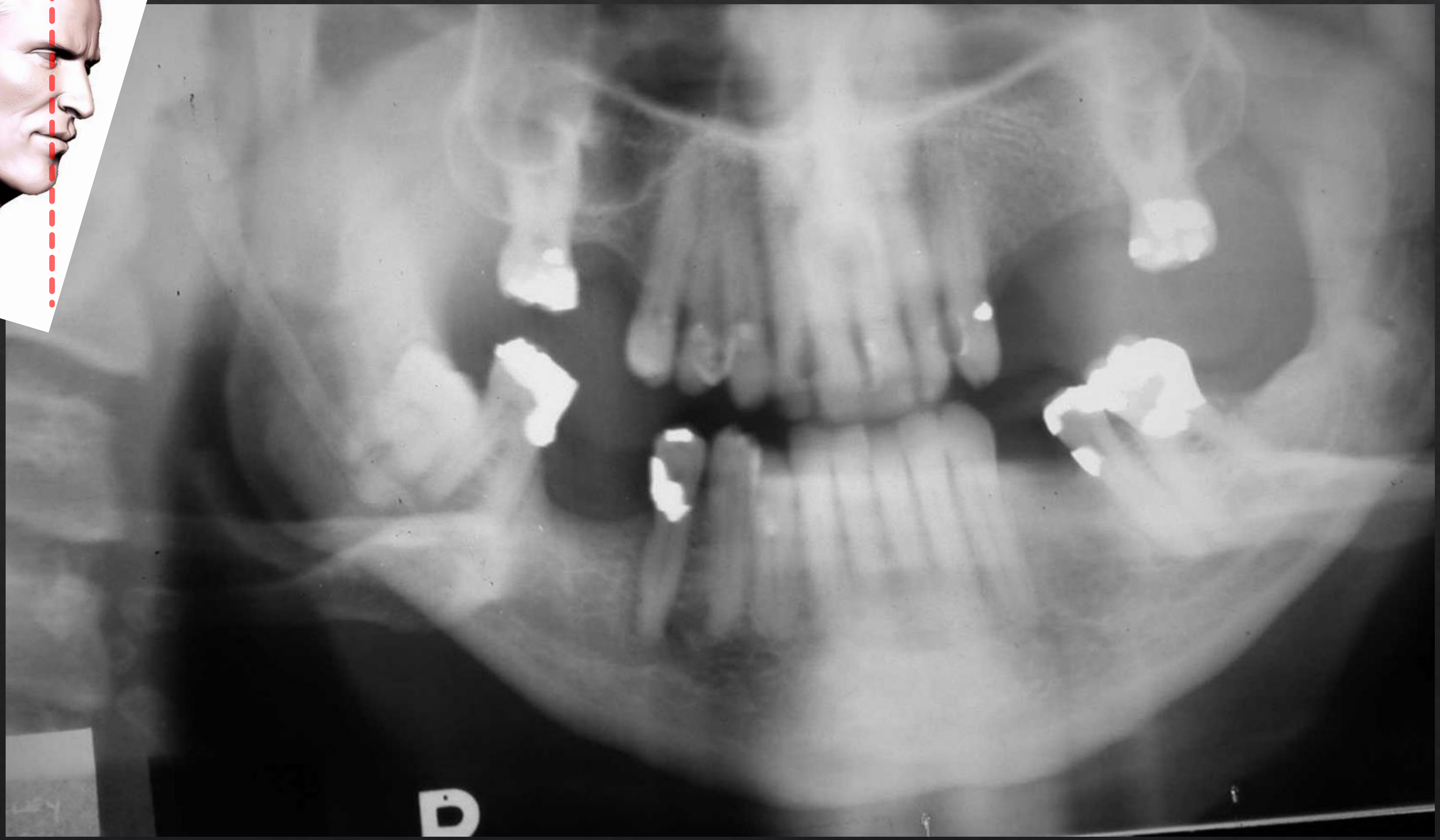


Patient positioning

Chin too high; upward overangulation

Patient positioning

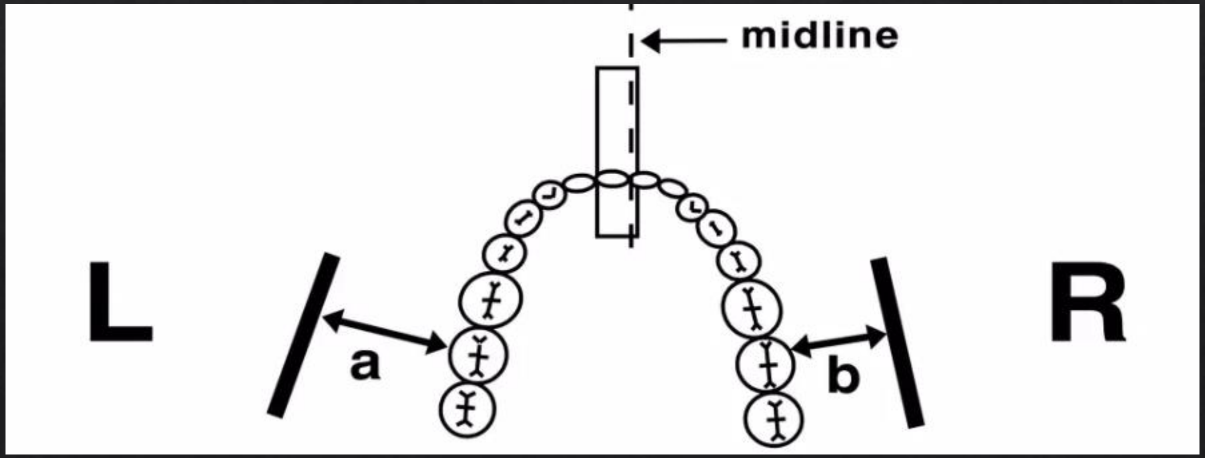
Chin too low; downward overangulation



Patient positioning

Rotation & shifting; left-right distortion

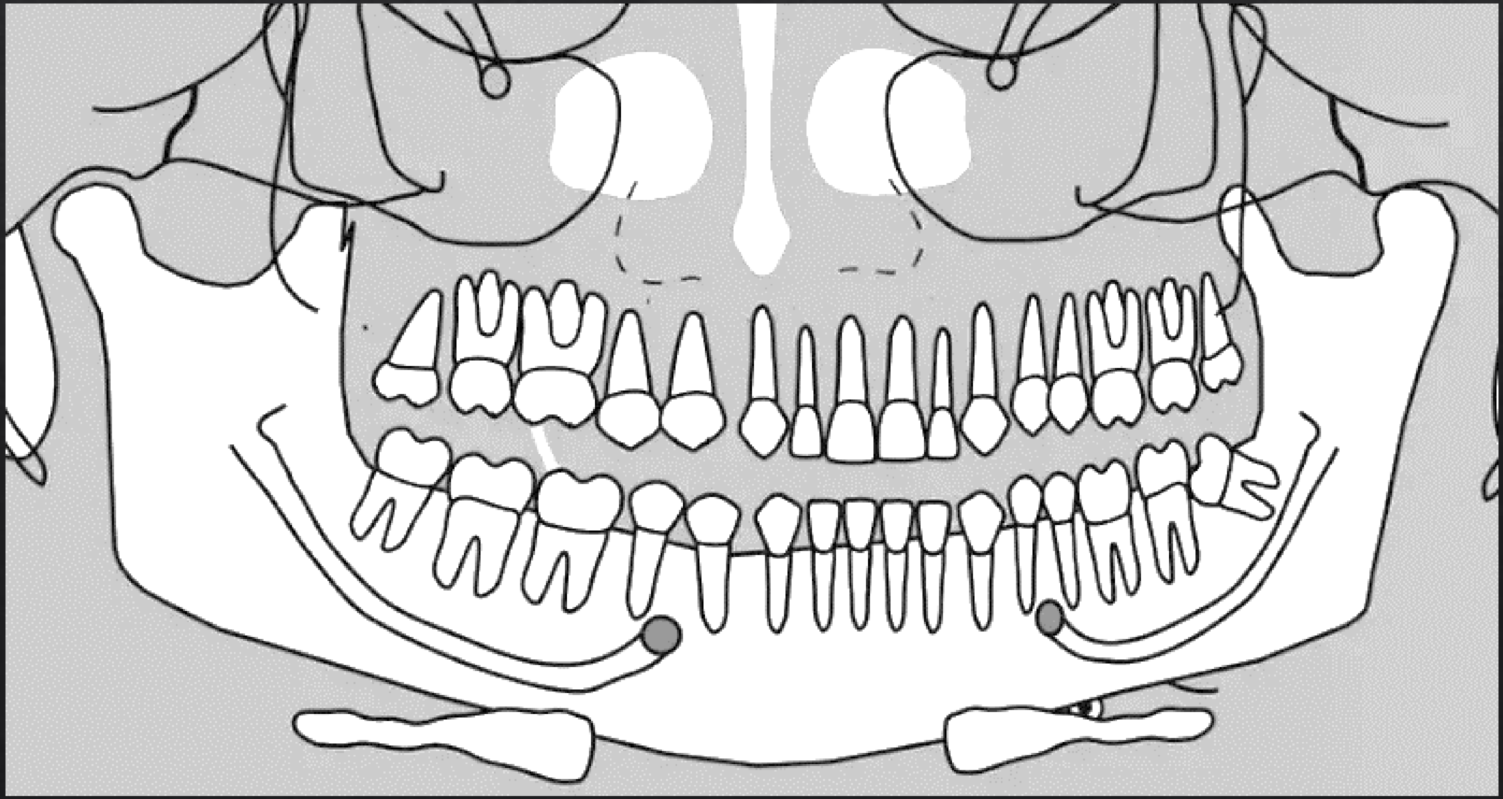
- If the head is turned slightly to the side (not centered in the bitestick), the structures on the one side will be closer to the film and the structures on the other side will be farther from the film,
- In the diagram below, the head was turned to the right and the teeth are closer to the film/sensor on that side



Patient positioning

Rotation & shifting; left-right distortion

- The teeth are smaller on the side to which the head is rotated.
- The teeth that are farther from the film/sensor are wider because there is increased magnification horizontally



Patient positioning

Rotation & shifting; left-right distortion

Wider side

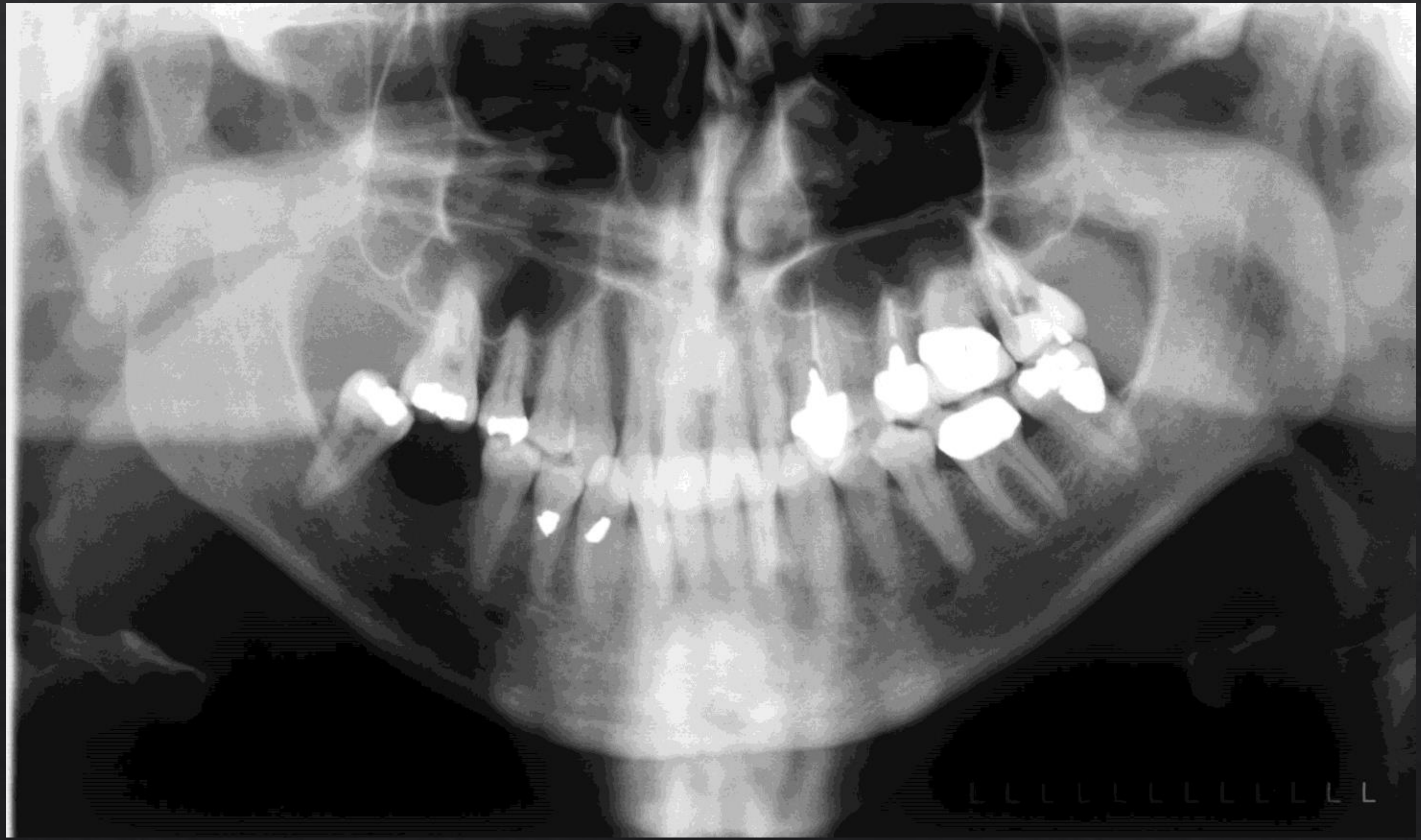
- Closer to the X-ray tube
- Buccal to the focal trough
- Away from the receptor
- Appears wide, blurred, magnified

Narrower side

- Closer to the image receptor
- Lingual to the focal trough
- Closer to the receptor
- Appears narrower and sharper

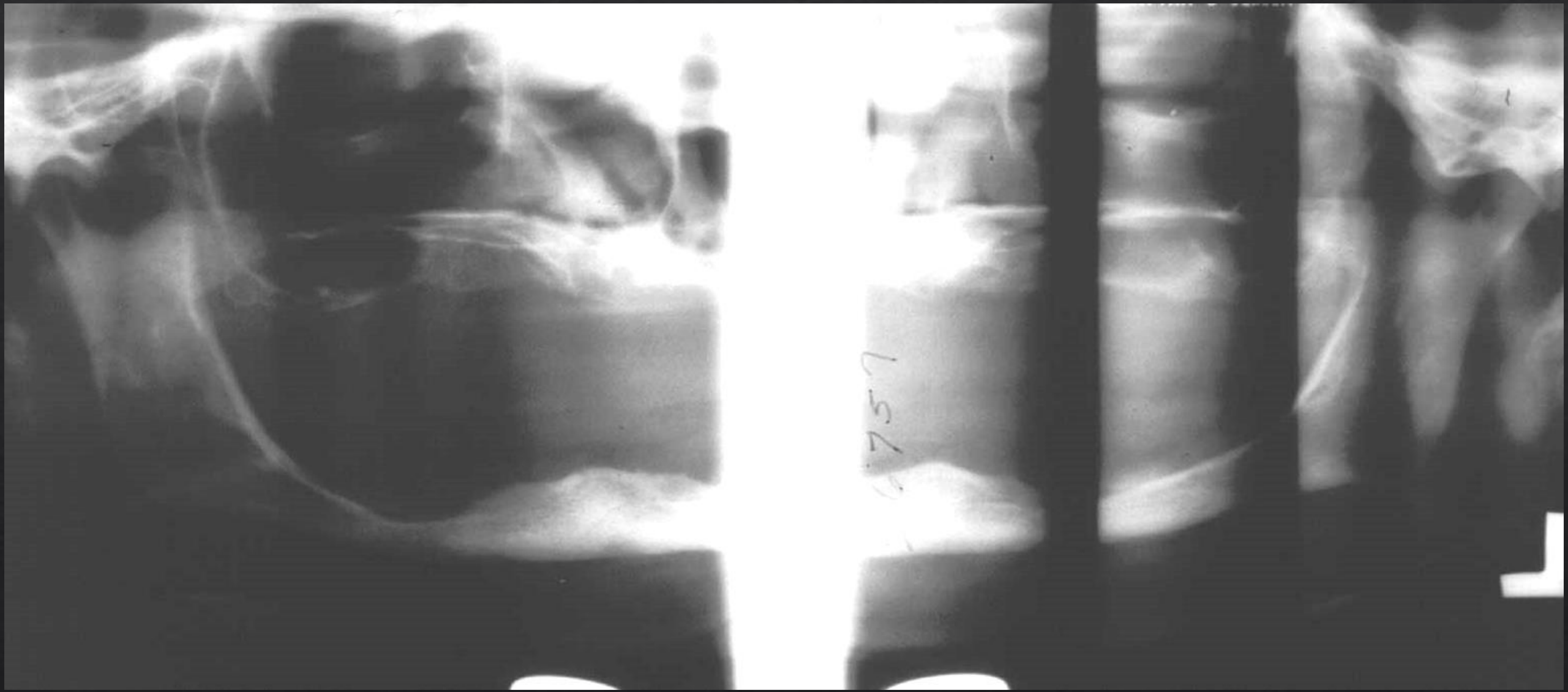
Patient positioning

Neck extension (e.g. patient standing too much back); cervical spine overlap



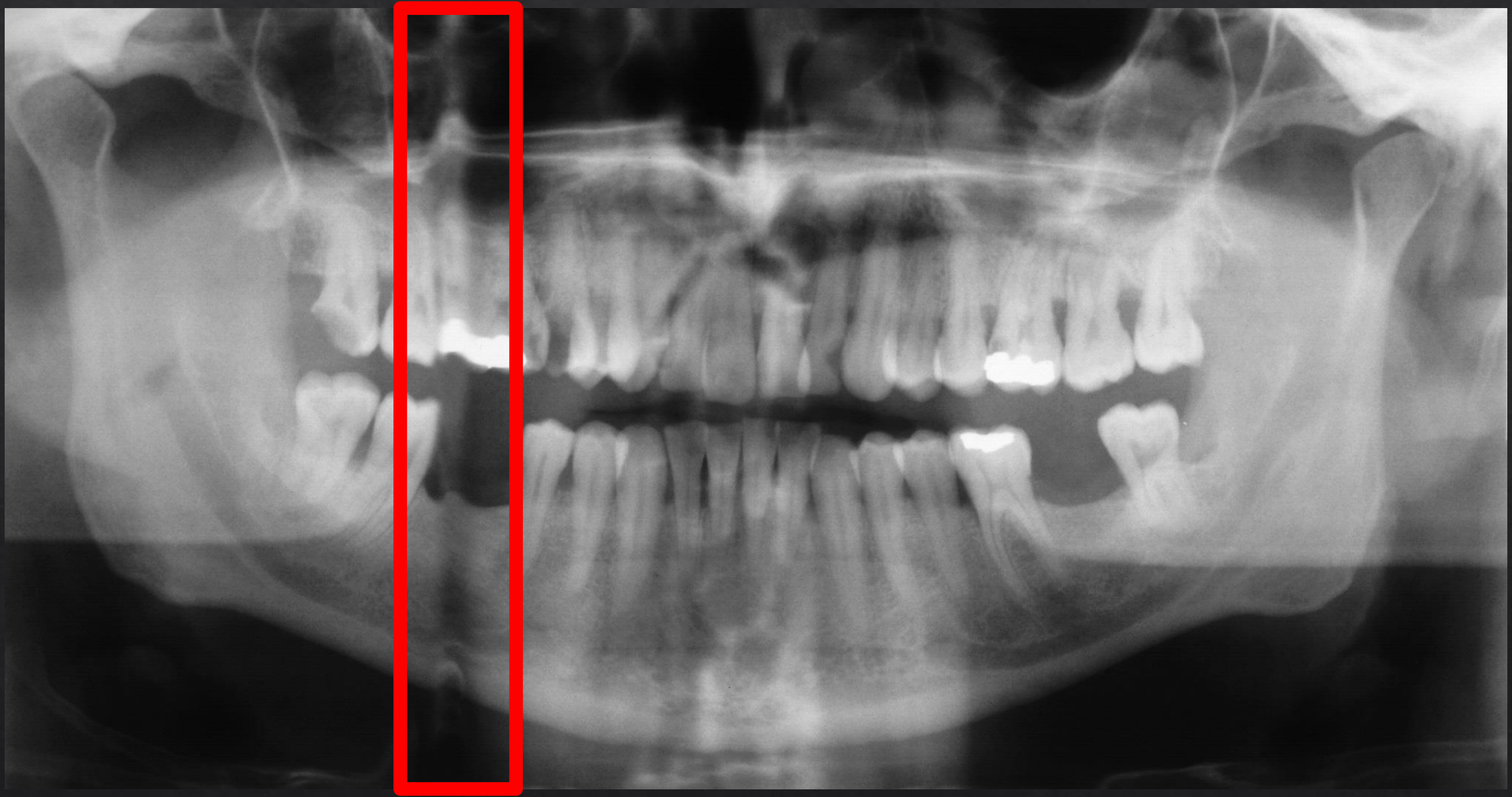
Patient positioning

Collision with shoulder



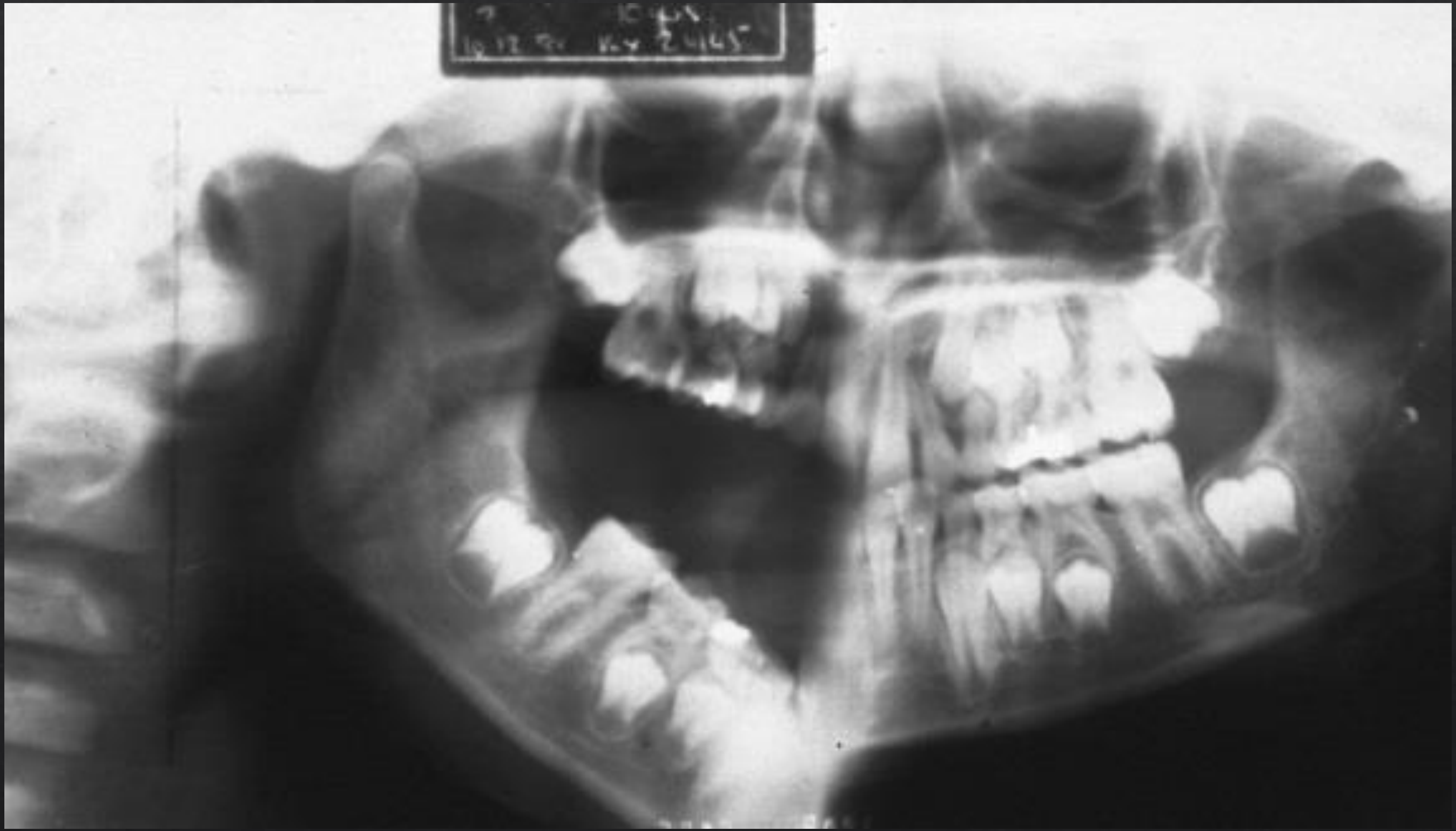
Patient positioning

Patient movement (slight)



Patient positioning

Patient movement (severe)

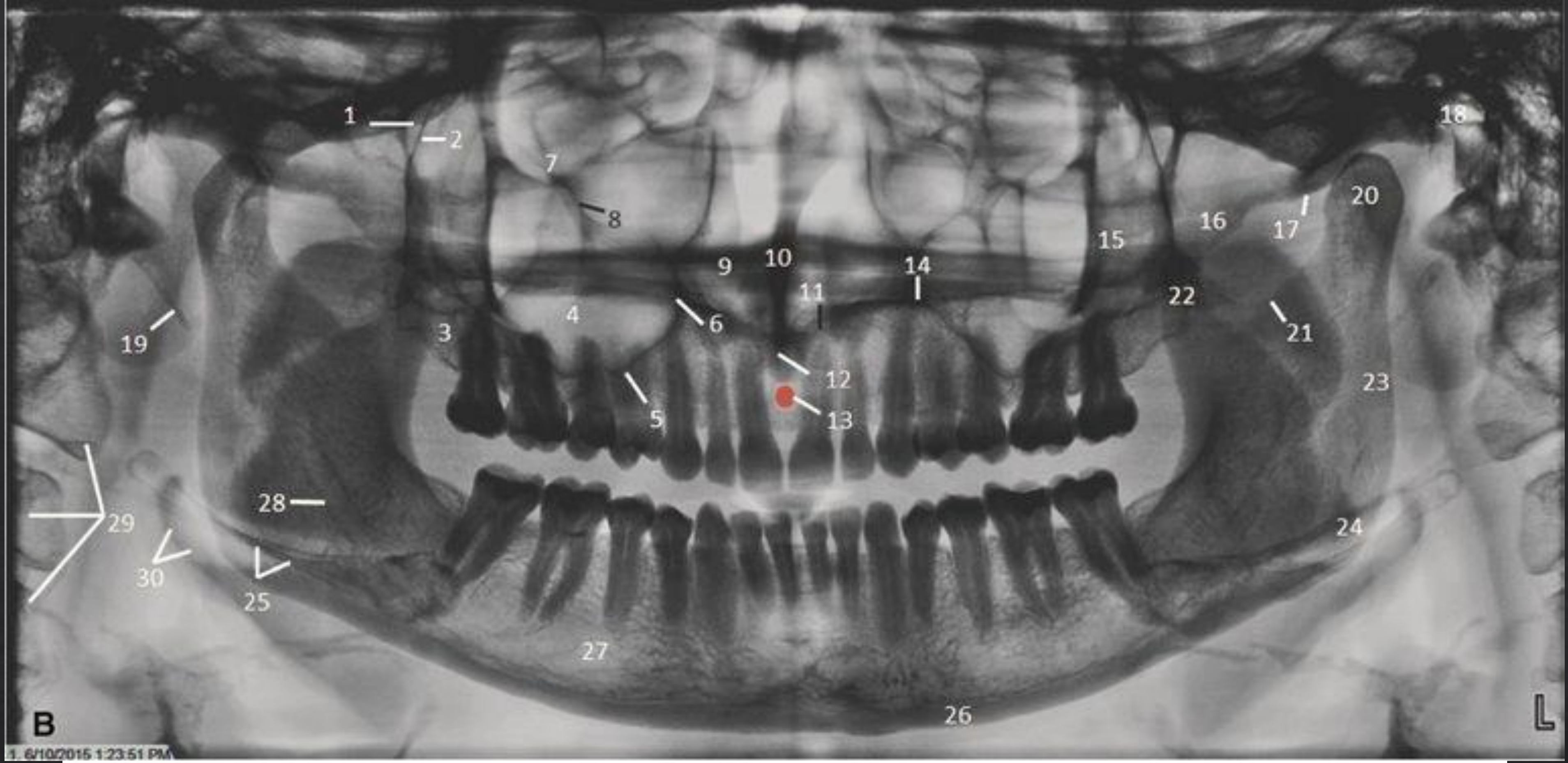


Patient positioning

Improper tongue placement

Patient positioning

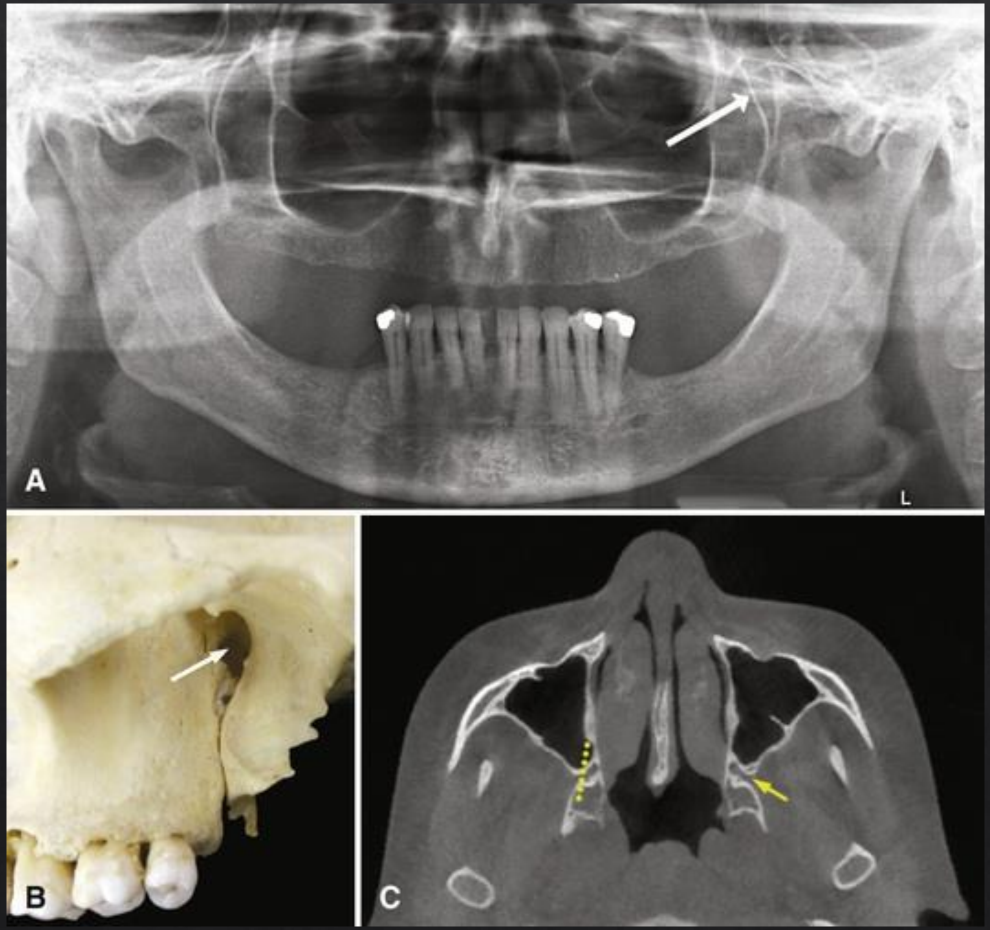
Lead apron artefact



1. Pterygomaxillary fissure
2. Posterior border of maxilla
3. Maxillary tuberosity
4. Maxillary sinus
5. Floor of the maxillary sinus
6. Medial border of the maxillary sinus/ lateral border of the nasal cavity
7. Floor of the orbit
8. Infraorbital canal
9. Nasal cavity
10. Nasal septum
11. Floor of the nasal cavity
12. Anterior nasal spine
13. Incisive foramen
14. Hard palate/floor of the nasal cavity
15. Zygomatic process of maxilla
16. Zygomatic arch
17. Articular eminence
18. External auditory meatus
19. Styloid process
20. Mandibular condyle
21. Sigmoid notch
22. Coronoid process
23. Posterior border of the ramus
24. Angle of the mandible
25. Hyoid bone
26. Inferior border of the mandible
27. Mental foramen
28. Mandibular canal
29. Cervical vertebrae
30. Epiglottis

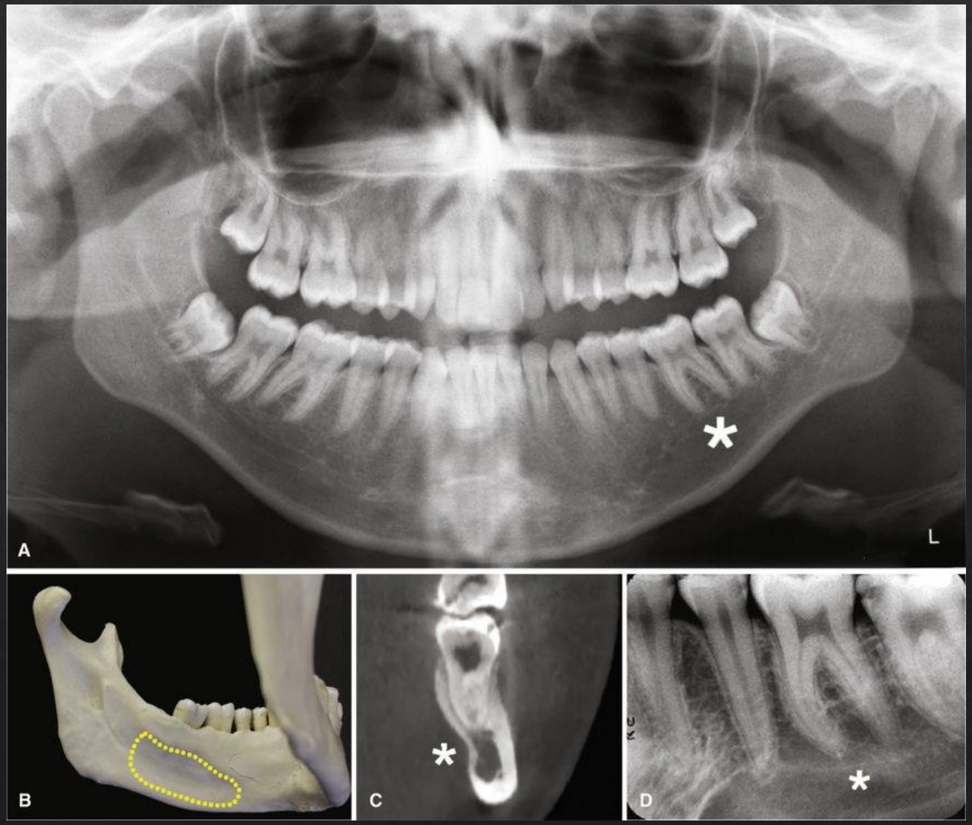
Pterygomaxillary fissure

- Space between the posterior surface of the maxilla and the anterior border of the pterygoid plates
- Inverted teardrop shape
- Maxillary sinus mucoceles and carcinomas destroy anterior border of PMF
- Contents: PSA?



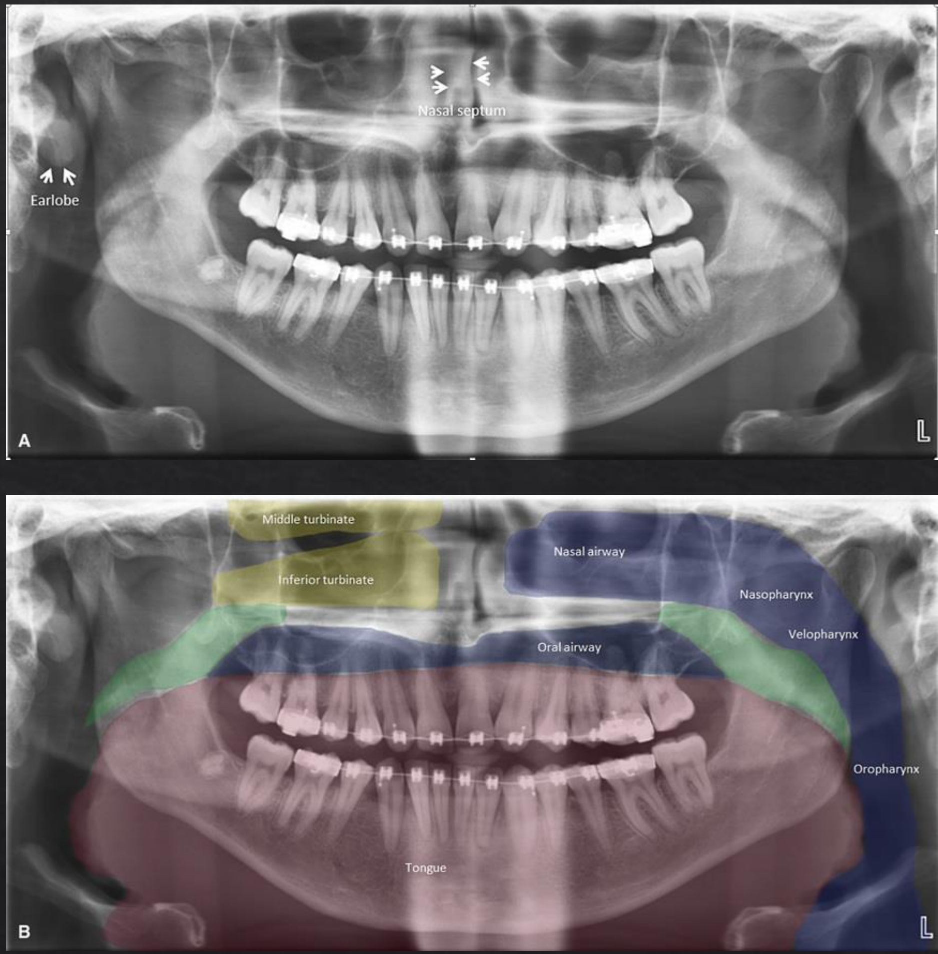
Lingual salivary gland depression

- A concavity often found on the posterior lingual surface of the mandible
- Bounded by the mylohyoid ridge superiorly and the inferior border of the mandibular body
- Lies in the region of the roots of the molars and premolars



Soft tissue images on a panoramic radiograph

- Panoramic image with an overlay indicating the components of the airway
- The nasal airway surrounds the turbinates
- The nasopharynx is posterior to the turbinates and above the level of the hard palate
- The velopharynx is posterior to the soft palate
- The oropharynx is below the uvula



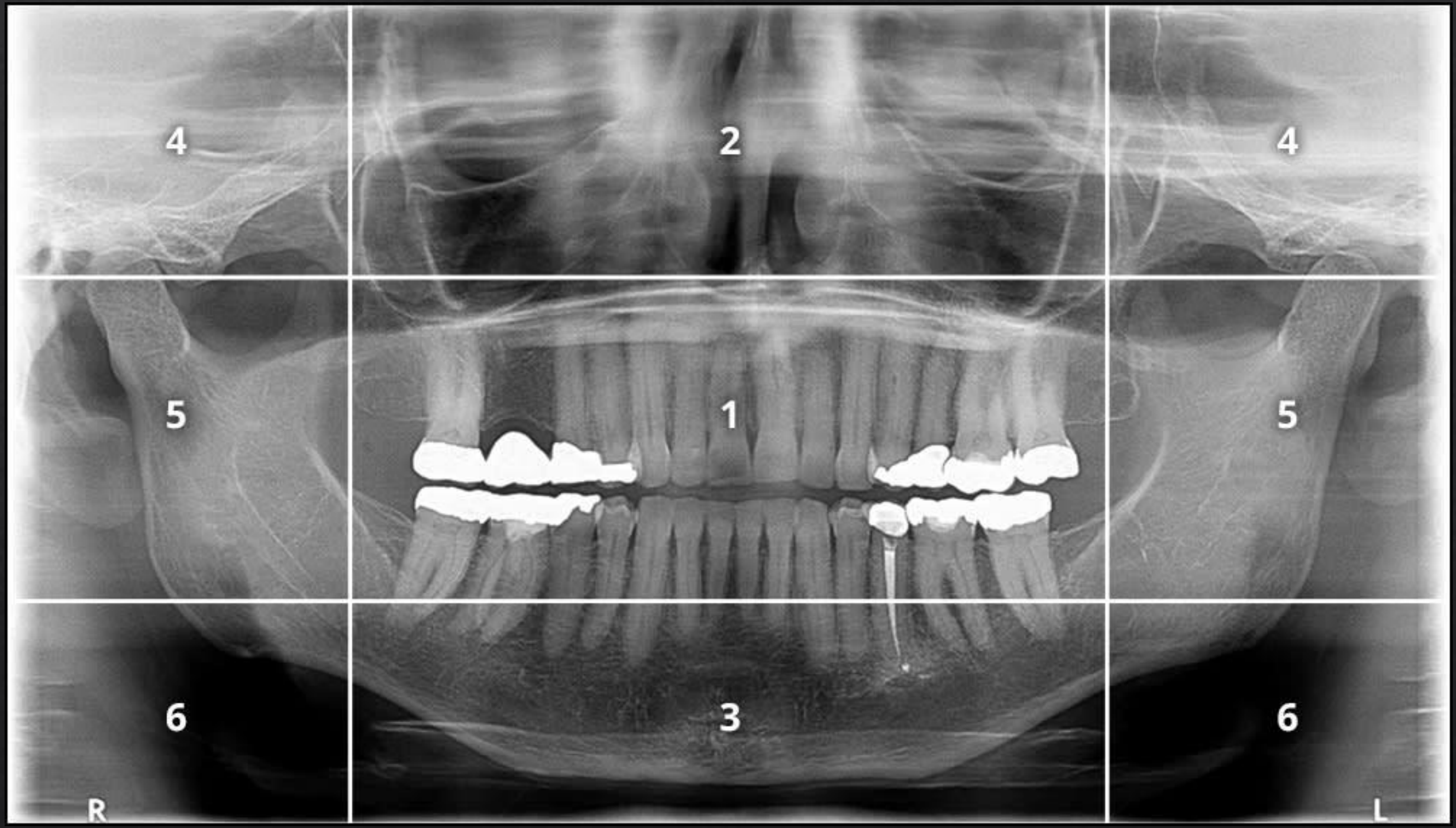
Interpreting Panoramic images: Method-1

- Entire image
- Global to local (leave teeth for last)
- Hard and soft tissues
- One component at the time

Check list:

- ✓ Technical quality
- ✓ Bone pathology
- ✓ TMJs
- ✓ Maxillary sinuses
- ✓ Soft tissue calcifications
- ✓ Teeth and periodontium

Interpreting Panoramic images: Method-2



Zones of Panoramic Image Interpretation

Zone 1: Dentition

- Teeth arranged with an upward smile-like curve
- Anterior teeth should not be too large or small
- Posterior teeth should be evenly sized without excessive overlap
- Apices/crowns of teeth should be visible

Zones of Panoramic Image Interpretation

Zone 2: Nose & Sinus

- Inferior turbinates and surrounding air spaces visible
- Soft tissue of nose should not be visible
- Shadow of hard palate will be seen in maxillary area
- Tongue should be in contact with hard palate

Zones of Panoramic Image Interpretation

Zone 3: Mandibular Body

- Inferior border of mandible should be continuous and smooth
- Ghost image of hyoid should not be visible
- Midline area should have proper proportions

Zone 4: Condyles

- Condyles should be centered within the area of the zone
- Condyles should be of equal size and on same horizontal plane

Zone 5: Ramus & Spine

- Ramus of mandible should be similar width bilaterally
- Spine, if seen, may be present as long as it doesn’t superimpose over the ramus
- If spine is present, the distance between the ramus and spine should be equivalent on both sides

Zone 6: Hyoid Bone

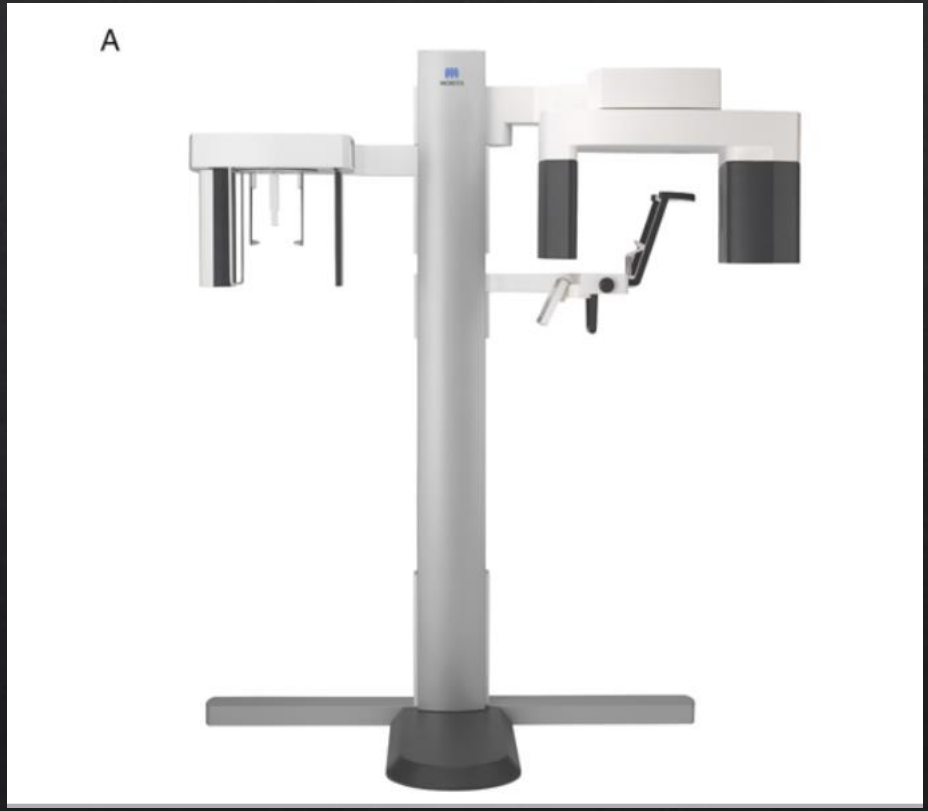
- Hyoid bone should appear as a bilateral, double image with equal proportions
- Hyoid image may touch the mandible, but should not spread across it

Error Visible on Panoramic Radiograph	Cause	Solution
Radiolucent artifact superimposed over maxillary apices — palatoglossal air space is visible.	Patient's tongue not placed against hard palate.	Instruct patient to place and hold tongue to the roof of the mouth for the entire exposure.
Radiolucent artifact superimposed over anterior teeth.	Patient's lips not closed around the bite block.	Instruct patient to close lips around the bite block.
Ghost image	Metal objects not removed from patient.	Remove all metal objects from the head and neck region prior to exposure.
Vertical radiopaque artifact in the center of the radiograph.	Cervical spine superimposed over anterior teeth — the patient is not standing or sitting as tall as possible.	Instruct patient to stand or sit as tall as possible, with shoulders relaxed. Instruct patient to hold handlebars with opposite hands or move feet forward toward the machine.
Radiopaque band or triangular opacity on the bottom edge of the radiograph.	A lead apron with a thyroid collar was used or the lead apron was draped too high on the patient.	Use a double-sided lead apron without a thyroid collar and drape the apron low around the neck and back area.
Anterior teeth blurry and narrow.	Patient is anterior to the focal trough.	Position the patient's anterior teeth in the bite-block's groove.
Anterior teeth blurry and wide.	Patient is posterior to the focal trough.	Position the patient's anterior teeth in the bite-block's groove.
One side of the radiograph is magnified.	Patient's head is leaning toward one side.	Position the midsagittal plane perpendicular to the floor.
Occlusal plane is flat or reversed. Maxillary anterior teeth are blurry and elongated. Condyles may not visible or located on the lateral sides of the radiograph.	Patient's chin is too high.	Position the Frankfort plane parallel to the floor.
Occlusal plane is exaggerated. Mandibular anterior teeth are blurry and foreshortened. Condyles are located at the top of the radiograph.	Patient's chin is too low.	Position the Frankfort plane parallel to the floor.

Panoramic machines

- Several manufacturers produce high-quality film-based and digital panoramic machines. Most of these units have the versatility to allow for adjustment of focal trough shape based on patient size (adult vs child) for panoramic image and produce tomographic cross-sectional images of selected areas of the facial skeleton.

- In addition to producing standard panoramic images of the jaws, some of these units have the capability of adjusting to patients of various sizes and making frontal and lateral images of TMJs, tomographic views through the sinuses, and cross-sectional views of the maxilla and mandible. These views are acquired by having special x-ray source and receptor movements programmed into the machine.

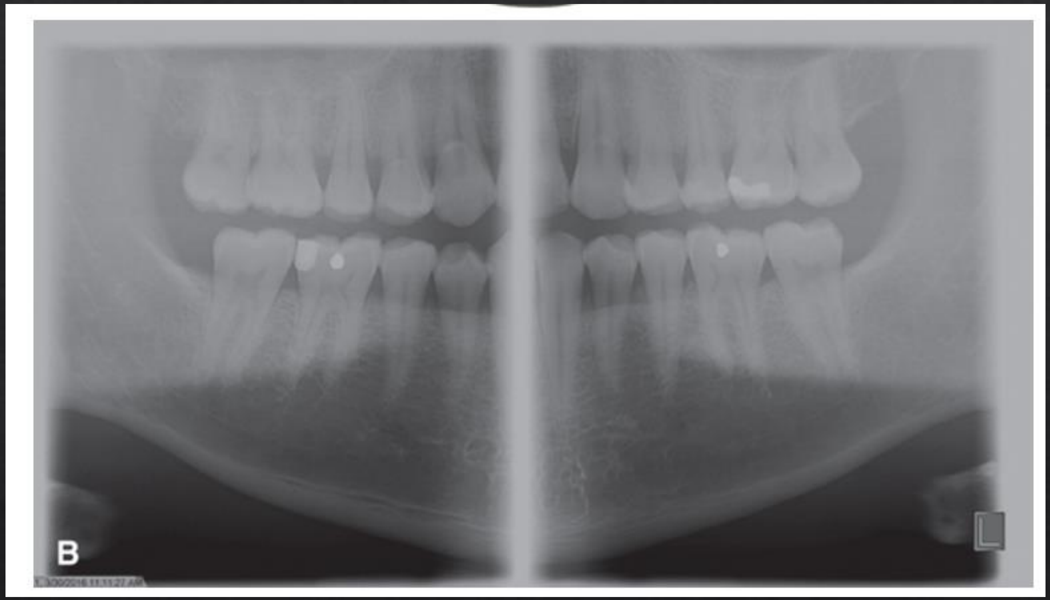


(A) Veraview X800, a multipurpose imaging device combining cephalometric, panoramic, and cone beam computed tomography imaging

Panoramic machines

"Extraoral bitewing" views are also offered by a few of the panoramic units that are designed to decrease overlap between the proximal surfaces of posterior teeth. Each machine also has the capability for adding on a cephalometric attachment to allow exposure of standardized skull views.

Some machines have the capability of automated exposure control; this is accomplished by measuring the amount of radiation passing through the patient's mandible during the initial part of the exposure and adjusting the imaging factors (kilovoltage peak, milliamperage, and speed of imaging movements) to obtain an optimally exposed image. Finally, all of these machines are available in CCD-digital configurations, and some have cone beam imaging capability



(B) "Extraoral bitewings" with open interproximal contacts in the posterior teeth as well as the periapical areas. Source: (A, Courtesy J. Morita Corporation, Kyoto, Japan; B, Courtesy Planmeca ProMax 3D, Planmeca Inc., Wood Dale, IL.)